# Nikon Corporation Type0016 vendor unique capabilities

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Nikon Corporation

## 1. Introduction

This document explains the vendor unique capabilities, which are used by Type0016module
(Type0016.md3, Type0016 module.bundle).

These definition values are defined in Maid3d1.h. Refer to the MAID 3.1 Specification for the details of capabilities.

NOTE) These unique capabilities may have different function at another module.

## 2. Supported camera

Type0016 module can control D5500, D5600 camera.

## 3. Vendor Unique Capabilities

The vendor unique capabilities that are used by Type0016 module are described as follows.

The under line shows default value.

### 3.1. ImageSize

This will set the size of image. (Shooting menu)

**Capability** kNkMAIDCapability_ImageSize

**Object types** Source

**ulType** kNkMAIDCapType_Enum
kNkMAIDArrayType_PackedString

**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set

**Data**

| | |
|---|---|
| L | L(6000*4000) |
| M | M(4496*3000) |
| S | S(2992*2000) |

The following cases, this Capability cannot be set into.

- When the value of Capability_CompressionLevel is “RAW”
- When the Capability_InfoDisplayErrStatus is True: ON (Error display).
- During movie file recording.

### 3.2. CompressionLevel

This will select the compression level of a picture. (Shooting Menu)

**Capability** kNkMAIDCapability_CompressionLevel

**Object types** Source

**ulType** kNkMAIDCapType_Enum
kNkMAIDArrayType_PackedString

**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set

**Data** JPEG Basic,
JPEG Normal,
JPEG Fine,
RAW,
RAW + JPEG Basic,
RAW + JPEG Normal,
RAW + JPEG Fine

The following cases, the module enumerates values which don’t include “RAW”

- When the value of Capability_ ExposureMode is EFFECTS (“Miniature Effect”,

"Selective Color", "Night Vision", "Toy Camera Effect", "Super Vivid", "Pop" or "Photo Illustration").

- When the value of Capability_HDRMode is except 0 (Off).

The following cases, this Capability is read only.

- When the Capability_InfoDisplayErrStatus is True: ON (Error display).
- During movie file recording.

### 3.3. WBMode

This will select the white balance mode. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_WBMode |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Enum<br>kNkMAIDArrayType_PackedString |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,<br>kNkMAIDCapOperation_Set |
| **Data** | Auto,<br>Incandescent,<br>Fluorescent,<br>Sunny,<br>Flash,<br>Shade,<br>Cloudy,<br>Measure,<br>Use Photo |

When the Capability_ExposureMode is set to "twilight" or "candle" in Scene modes, "K" is displayed on the camera body. However, the value of this Capability “Auto” returns.

When the Capability_HDRMode is except “0: Off” and the value of this Capability is “Measure”, “Measure” is not enumerated.

The following cases, this Capability cannot be set into.

- When the Capability_ExposureMode is Scene Modes or Special Effects Modes (5: [Scene Modes] Auto, 13: [Scene Modes] Auto (flash off), 14: [Scene Modes] SCENE, 17: [Special Effects] EFFECTS).
- When the Capability_InfoDisplayErrStatus is True: ON (Error display).
- During movie file recording.

### 3.4. Sensitivity

This will select the sensitivity of camera (Shooting menu)

**Capability** kNkMAIDCapability_Sensitivity

**Object types** Source

**ulType** kNkMAIDCapType_Enum
kNkMAIDArrayType_PackedString

**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set

**Data**

| |
|---|
| Auto |
| 100 |
| 125 |
| 160 |
| 200 |
| 250 |
| 320 |
| 400 |
| 500 |
| 640 |
| 800 |
| 1000 |
| 1250 |
| 1600 |
| 2000 |
| 2500 |
| 3200 |
| 4000 |
| 5000 |
| 6400 |
| 8000 |
| 10000 |
| 12800 |
| 16000 |
| 20000 |
| 25600 |

When Capability_ExposureMode is set to “Program mode” or “Aperture priority” or “Speed priority”, and “Manual”, “Auto” is not enumerated.

When Capability_ExposureMode is set to “Auto” or “Auto (flash off)” or “Night vision” of Special Effects Modes, the current value is fixed at “Auto”.

By setting Capability_ExposureMode, value range that can be set is limited as follows.

| ExposureMode | ISO value range |
|---|---|
| P,S,A,M | 100～25600 |
| Auto, Auto(flash off), Night Vision( EFFECTS) | Auto |
| Other | Auto, 100～25600 |

The following cases, this Capability cannot be set into.

- When Capability_ExposureMode is set to “Auto” , “Auto (flash off)” or “Night vision” of Special Effects Modes.
- When the Capability_InfoDisplayErrStatus is True: ON (Error display).

### 3.5. ResetMenuBank

This will reset the shooting menu. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_ResetMenuBank |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Process |
| **ulOperations** | kNkMAIDCapOperation_Start |
| **Data** | None |

During movie file recording, this capability cannot be performed.

### 3.6. WBTuneAuto

This will set the white balance adjustment when the WBMode is “Auto”. (Shooting menu)

**Capability** kNkMAIDCapability_WBTuneAuto

**Object types** Source

**ulType** kNkMAIDCapType_Range

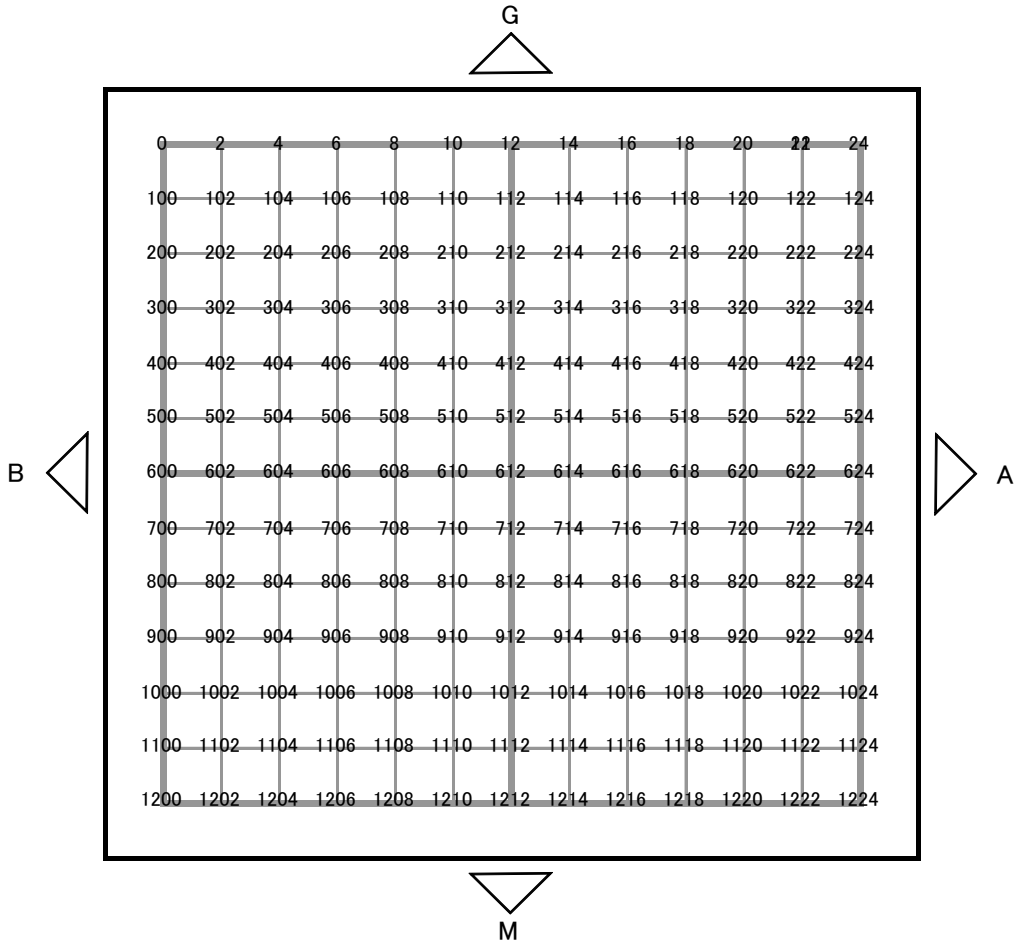
**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

**Data** 0 to 1224step=1 (Default: 612)

The relationship between white balance adjustment value and the coordinates is shown infollowing figure.

A-B: 0.5step

G-M: 0.25step



The following cases, this Capability cannot be set into.

- When the Capability_ExposureMode is Scene Modes or Special Effects Modes (5: [Scene Modes] Auto, 13: [Scene Modes] Auto(flash off), 14: [Scene Modes] SCENE 17: [Special Effects] EFFECTS).
- During movie file recording.

### 3.7. WBTuneIncandescent

This will set the white balance adjustment when the WBMode is “Incandescent”. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_WBTuneIncandescent |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Range |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set |
| **Data** | 0 to 1224 step=1 (Default: 612) |

The relationship between white balance adjustment value and the coordinates is the same as the figure of Capability_WBTuneAuto.

The following cases, this Capability cannot be set into.

- When the Capability_ExposureMode is Scene Modes or Special Effects Modes (5: [Scene Modes] Auto, 13: [Scene Modes] Auto(flash off), 14: [Scene Modes] SCENE 17: [Special Effects] EFFECTS).
- During movie file recording.

### 3.8. WBFluorescentType

This will set the fluorescent type when the WBMode is “Fluorescent”. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_WBFluorescentType |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault |
| **Data** | one of eNkWBFluorescentType<br>0: Sodium-vapor lamps<br>1: Warm-white fluorescent<br>2: White fluorescent<br>3: Cool-white fluorescent<br>4: Day white fluorescent<br>5: Daylight fluorescent<br>6: High temp.mercury-vapor |

The following cases, this Capability cannot be set into.

- When the Capability_ExposureMode is Scene Modes or Special Effects Modes.
- During movie file recording.

### 3.9. WBTuneFluorescent

This will set the white balance adjustment when the WBMode is “Fluorescent”. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_WBTuneFluorescent |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Range |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set |
| **Data** | 0 to 1224step=1　(Default: 612) |

The relationship between white balance adjustment value and the coordinates is the same as the figure of Capability_WBTuneAuto.

The following cases, this Capability cannot be set into.

- When the Capability_ExposureMode is Scene Modes or Special Effects Modes.
- During movie file recording

### 3.10. WBTuneSunny

This will set the white balance adjustment when the WBMode is “Sunny”. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_WBTuneSunny |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Range |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set |
| **Data** | 0 to 1224step=1　(Default: 612) |

The relationship between white balance adjustment value and the coordinates is the same as the figure of Capability_WBTuneAuto.

The following cases, this Capability cannot be set into.

- When the Capability_ExposureMode is Scene Modes or Special Effects Modes.
- During movie file recording.

### 3.11. WBTuneFlash

This will set the white balance adjustment when the WBMode is “Flash”. (Shooting menu)

**Capability** kNkMAIDCapability_WBTuneFlash
**Object types** Source
**ulType** kNkMAIDCapType_Range
**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
**Data** 0 to 1224 step=1 (Default: 612)

The relationship between white balance adjustment value and the coordinates is the same as the figure of Capability_WBTuneAuto.

The following cases, this Capability cannot be set into.

- When the Capability_ExposureMode is Scene Modes or Special Effects Modes.
- During movie file recording.

### 3.12. WBTuneShade

This will set the white balance adjustment when the WBMode is “Shade”. (Shooting menu)

**Capability** kNkMAIDCapability_WBTuneShade
**Object types** Source
**ulType** kNkMAIDCapType_Range
**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
**Data** 0 to 1224step=1 (Default: 612)

The relationship between white balance adjustment value and the coordinates is the same as the figure of Capability_WBTuneAuto.

The following cases, this Capability cannot be set into.

- When the Capability_ExposureMode is Scene Modes or Special Effects Modes.
- During movie file recording

### 3.13. WBTuneCloudy

This will set the white balance adjustment when the WBMode is “Cloudy”. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_WBTuneCloudy |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Range |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set |
| **Data** | 0 to 1224step=1　(Default: 612) |

The relationship between white balance adjustment value and the coordinates is the same as the figure of Capability_WBTuneAuto.

The following cases, this Capability cannot be set into.

- When the Capability_ExposureMode is Scene Modes or Special Effects Modes.
- During movie file recording.

### 3.14. WBPresetNumber

This will set the preset number referenced by the Capability_WBGainRed, Capability_WBGainBlue. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_WBPresetNumber |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Enum<br>kNkMAIDArrayType_PackedString |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,<br>kNkMAIDCapOperation_Set |
| **Data** | Measure, Use photo |

### 3.15. WBPresetData

This will set the white balance preset data to the camera. (Shooting menu)

**Capability** kNkMAIDCapability_WBPresetData

**Object types** Source

**ulType** kNkMAIDCapType_Generic

**ulOperations** kNkMAIDCapOperation_Set

**Data** pointer to NkMAIDWBPresetData structure

```
typedef struct tagNkMAIDWBPresetData
{
ULONG   ulPresetNumber;------ (This member is not used)
ULONG   ulPresetGain;---------- gain value
ULONG   ulThumbnailSize;----- the thumbnail size set to
"pThumbnailData"
ULONG ulThumbnailRotate;--- (This member is not used)
void*   pThumbnailData;--------- the pointer to the thumbnail data to
be set.
} NkMAIDWBPresetData, FAR* LPNkMAIDWBPresetData;
```

The client must to set all the member of "NkMAIDWBPresetData" structure without "ulThumbnailRotate". The data which is set is saved to "d-1" data area.

The member "ulThumbnailSize" and "pThumbnailData" of "NkMAIDWBPresetData" structure is used only for kNkMAIDCapOperation_Set.

The red gain value is set to the upper 2 bytes, the blue gain value is set to the lower 2 bytes of "ulPresetGain".

The both of red and blue gain value is expressed by the 8.8 format fixed-point number. (e.g. 1.5 → gain value: 0x0180) The range of gain value is $0 \leq$ gain value $< 8$ (0x0000 -0x07FF).

The thumbnail data set to "pThumbnailData" must be matching the following requirement.

1. Theimage data is JPEG.
2. The size of image is 160 x 120.
3. The quality of image is Fine(1/4 compressed).
4. The size of image is below 13440 bytes.
5. The image cannot include the tag except the following table.

When during movie file recording, this capability cannot be set into.

| | |
|---|---|
| SOI | Start Of Image |
| DQT | Define Quantization Table |
| DHT | Define Huffman Table |
| SOF | Start of Frame |
| SOS | Start of Scan |
| | Entropy Coded Data<br>(JPEG data) |
| EOI | End Of Image |

### 3.16. WBGainRed

This will get the gain red of white balance preset data selected by the Capability_WBPresetNumber. (Shooting menu)

**Capability** kNkMAIDCapability_WBGainRed
**Object types** Source
**ulType** kNkMAIDCapType_Range
**ulOperations** kNkMAIDCapOperation_Get
**Data** Min: 0 Max: 7.9661 (2047/256)
Step: 0.0039 (1/256) (Default: 1)

### 3.17. WBGainBlue

This will get the gain blue of white balance preset data selected by the Capability_WBPresetNumber. (Shooting menu)

**Capability** kNkMAIDCapability_WBGainBlue
**Object types** Source
**ulType** kNkMAIDCapType_Range
**ulOperations** kNkMAIDCapOperation_Get
**Data** Min: 0 Max: 7.9661 (2047/256)
Step: 0.0039 (1/256) (Default: 1)

### 3.18. ImageColorSpace

This will set color space. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_ImageColorSpace |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get,<br>kNkMAIDCapOperation_Set,<br>kNkMAIDCapOperation_GetDefault |
| **Data** | one of eNkMAIDImageColorSpace<br>0: sRGB,<br>1: AdobeRGB |

When during movie file recording, this Capability cannot be set into.

### 3.19. IsoControl

This will set whether auto sensitivity control is used when you take a picture. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_IsoControl |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Boolean |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set,<br>kNkMAIDCapOperation_GetDefault |
| **Data** | True: used False: not used |

When this capability value is set to True, ISO is controlled automatically by the camera in taking picture.

The following cases, this Capability cannot be set into.

- When the Capability_ExposureMode is Scene Modes or Special Effects Modes.
- During movie file recording

### 3.20. NoiseReduction

This will set whether noise reduction is used or not used. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_NoiseReduction |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Boolean |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault |
| **Data** | True: used False: not used |

The following cases, this Capability cannot be set into.

- When the Capability_ExposureMode is Special Effects Modes(Night Vision).
- During movie file recording

### 3.21. NoiseReductionHighISO

This will set whether noise reduction is used or not used when high ISO. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_NoiseReductionHighISO |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault |
| **Data** | one of eNkMAIDNoiseReductionHighISO<br>0: OFF<br>1: ON (Normal)<br>2: ON (High)<br>3: ON (Low) |

The following cases, this Capability cannot be set into.

- When the Capability_ExposureMode is Special Effects Modes(Night Vision).
- During movie file recording

### 3.22. CompressRAWBitMode

This will select bit depth for RAW(NEF). (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_CompressRAWBitMode |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set |
| **Data** | one of eNkMAIDCompressRAWBitMode<br>0: 12-bit<br>1: 14-bit |

When during movie file recording, this Capability cannot be set into.

### 3.23. PictureControl

This will select Picture Control. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_PictureControl |
| **ulType** | kNkMAIDCapType_Enum<br>kNkMAIDArrayType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set |
| **Data** | one of eNkMAIDPictureControl<br>0: Undefined Picture Control<br>1: Standard<br>2: Neutral<br>3: Vivid<br>4: Monochrome<br>5: Portrait<br>6: Landscape<br>7: Flat<br>201 - 209: Custom Picture Control 1 - 9 |

This capability shows the current selected Picture Control.

When the client sends kNkMAIDCapOperation_GetArray, the module returns the all Picture control enumeration value including unused Custom Picture Control.

The client can know whether the Picture Control is used or not by checking "CustomFlag" in Picture Control Data format.(see Capability_PictureControldata)

When the client sends kNkMAIDCapOperation_Set with unused Picture Control, the module returns kNkMAIDResult_DeviceBusy.

When the Picture Control selected currently is changed, kNkMAIDEvent_CapChangeValueOnly is issued about this capability. And when the content of Picture Control data is changed, kNkMAIDEvent_CapChange is issued about Capability_ChangedPictureControl.

The following cases, this Capability cannot be set into.

- When the Capability_ExposureMode is Scene Modes or Special Effects Modes.
- During movie file recording
- When the Capability_InfoDisplayErrStatus is True: ON (Error display).

### 3.24. ChangedPictureControl

This will enumerate the Picture Control item, which is the content, is changed.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_ChangedPictureControl |
| **ulType** | kNkMAIDCapType_Enum |
| | kNkMAIDArrayType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, |
| **Data** | one of eNkMAIDPictureControl |

When the Picture Control is modified by the following factor, this capability reports the list of modified Picture Control to the client by kNkMAIDCapOperation_GetArray.

- The Picture Control was changed by edit.
- The Custom Picture Control was saved.
- The Custom Picture Control was deleted.
- The Custom Picture Control was renamed.

The current value of this capability shows the last modified Picture Control.

After the client gets the list of modified Picture Control by kNkMAIDCapOperation_GetArray, the module resets the enumeration data and the current value of this capability will be reset to 0, and the list of modified Picture Control will be deleted.

When the Picture Control is reset, kNkMAIDEvent_CapChange is not issued.

### 3.25. PictureControlData

This will set the assigned Picture Control data of the first generation. And this will get Picture Control data of the second generation. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_PictureControlData |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Generic |
| **ulOperations** | kNkMAIDCapOperation_Set, kNkMAIDCapOperation_Get<br>kNkMAIDCapOperation_GetDefault |
| **Data** | pointer to NkMAIDPicCtrlData structure<br>typedef struct tagNkMAIDPicCtrlData<br>{<br>ULONG ulPicCtrlItem;------ The target Picture Control<br>ULONG ulSize;------The size of Picture Control data (Max: 610 bytes)<br>bool bModifiedFlag; --------Modification flag(false: initial registration, true: edit)<br>void* pData;---------- The pointer of Picture Control data.<br>} NkMAIDPicCtrlData, FAR* LPNkMAIDPicCtrlData; |

The range of value sets to "ulPicCtrlItem" is enumerated by Capability_PictureControl.

The following cases, this Capability is read only.

- During movie file recording.

**[In case of Set]**

In this Capability, Picture Control data of the first generation will be always set.

When the client sends kNkMAIDCapOperation_Set, the client must set the all the member of NkMAIDPicCtrlData.

If "bModifiedFlag" is false (initial registration), the module updates the current value and default value of Picture Control, by the content of "pData". If "bModifiedFlag" is true (edit), the module updates the current value of Picture Control only, by the content of "pData". The limitations at Set are as follows.

- If "ulPicCtrlItem" is Standard(1), Neutral(2), Vivid(3), Monochrome(4), Portrait(5),Landscape(6), bModifiedFlag must be set to true(edit).
- If "ulPicCtrlItem" is Custom Picture Control(201 - 209), the "CustomFlag" of Picture Control data must be set to custom (1).
- □"RegistrationName" will not be used when "ulPicCtrlItem" is Standard(1),

Neutral(2),Vivid(3), Monochrome(4), Portrait(5), Landscape(6).

- If “ulPicCtrlItem” is Neutral(2), Custom Picture Control(201 - 209), the
- “QuickAdjustFlag” of Picture Control data must be set to invalid (0).
- □If “ulPicCtrlItem” is Monochrome(4), “MonochromeFlag” of Picture Control data must be set to monochrome (1). If “ulPicCtrlItem” is not Monochrome(4), “MonochromeFlag” of Picture Control data must be set to color(0).
- When “MonochromeFlag” is changed, bModifiedFlag must be set to false(initial registration).
- If the “QuickAdjustFlag” of Picture Control data is valid (1), the camera determines each setting by referring ”QuickAdjust” of Picture Control data, and does not refer the other settings. If “QuickAdjustFlag” of Picture Control data is invalid (0), the camera determines each setting by referring the other settings, and does not refer ”QuickAdjust” of Picture Control data.
- If “CustomCurveFlag” of picture control data is used (1), the client have to set Custom Picture Control(201 - 209) to “ulPicCtrlItem”.

**[In case of Get]**

This Capability will get only Picture Control data of the second generation.

When the client sends kNkMAIDCapOperation_Get, the client must set the maximum Picture Control data size, 610, to "ulSize", and set the allocation space for 610 bytes to ”pData”.

The module sets the size of the picture control data actually set to “pData” to “ulSize” when succeeding in acquisition.

It is possible to get Picture Control data about unused Picture Control data.

The client can know whether the Picture Control data is used or not by referring “CustomFlag”. The format of the Picture Control data is shown below.

[Color] First generation

| Field | Size (Byte) | Data |
|---|---|---|
| PicCtrlItem | 1 | type of Picture Control<br>1: Standard<br>2: Neutral<br>3: Vivid<br>4: Monochrome<br>5: Portrait<br>6: Landscape<br>In case of Custom Picture Control, set the base Picture Control. |
| MonochromeFlag | 1 | Monochrome Flag<br>0: color<br>1: monochrome |
| CustomFlag | 1 | Custom Flag<br>0 : Standard<br>1 : Custom<br>2 : Unused custom |
| RegistrationName | 20 | Registration name of Picture Control<br>The string data is 20 byte fixation, and null terminated.<br>(19 characters in actual.) |
| QuickAdjustFlag | 1 | Quick Adjust Flag<br>0: invalid<br>1: valid<br>In case of ulPicCtrlItem of NkMAIDPicCtrlData is Neutral or Custom Picture Control, it is 0 fixation. |
| QuickAdjust | 1 | Quick Adjust value<br>-2 to +2 |
| Saturation | 1 | Saturation<br>-3 to +3<br>-128 is Auto |
| Hue | 1 | Hue<br>-3 to +3 |
| Sharpening | 1 | Sharpening<br>0 to 9<br>-128 is Auto |

| | | |
|---|---|---|
| Contrast | 1 | Contrast<br>-3 to +3 -128 is Auto<br>If CustomCurveData is used, this setting is not referred. |
| Brightness | 1 | Brightness<br>-1 to +1<br>If CustomCurveData is used, this setting is not referred. |
| CustomCurveFlag | 1 | Custom Curve Flag<br>0: No Custom Curve<br>1: Custom Curve used |
| CustomCurveData | 578 | Custom Curve Data<br>This data is not added when there is no Custom Curb.<br>[Header] 64 byte + [LUT] 257x 2 byte = 578 byte<br>Refer to "LUT format" for details. |

### [Monochrome] First generation

| Field | Size (Byte) | Data |
|---|---|---|
| PicCtrlItem | 1 | type of Picture Control<br>1: Standard<br>2: Neutral<br>3: Vivid<br>4: Monochrome<br>5: Portrait<br>6: Landscape<br>In case of Custom Picture Control, set the base Picture Control. |
| MonochromeFlag | 1 | Monochrome Flag<br>0: color<br>1: monochrome |
| CustomFlag | 1 | Custom Flag<br>0 : Standard<br>1 : Custom<br>2 : Unused custom |
| RegistrationName | 20 | Registration name of Picture Control<br>The string data is 20 byte fixation, and null terminated.<br>(19 characters in actual.) |
| FilterEffects | 1 | Filter Effect |

| | | |
|---|---|---|
| | | 0: None<br>1: Yellow<br>2: Orange<br>3: Red<br>4: Green |
| Toning | 1 | Toning(ToneColor)<br>0:B&W<br>1:Sepia<br>2:Cyanotype<br>3:Red<br>4:Yellow<br>5:Green<br>6:Blue Green<br>7:Blue<br>8:Purple Blue<br>9:Red Purple |
| ToningDensity | 1 | Toning(Level)(1Step)<br>1 to 7 |
| Reserve | 1 | Vacant |
| Sharpening | 1 | Sharpening(1Step)<br>0 to 9<br>-128 is Auto |
| Contrast | 1 | Contrast(1Step)<br>-3 to +3<br>-128 is Auto<br>If CustomCurveData is used, this setting is not referred. |
| Brightness | 1 | Brightness(1Step)<br>-1 to +1<br>If CustomCurveData is used, this setting is not referred. |
| CustomCurveFlag | 1 | Custom Curve Flag<br>0 : No Custom Curve<br>1 : Custom Curve used |
| CustomCurveData | 578 | Custom Curve Data<br>This data is not added when there is no Custom Curb.<br>[Header] 64 byte + [LUT] 257x 2 byte = 578 byte<br>Refer to "LUT format" for details. |

[Color] Second generation

| Field | Size (Byte) | Data |
|---|---|---|
| PicCtrlItem | 1 | type of Picture Control<br>1: Standard<br>2: Neutral<br>3: Vivid<br>4: Monochrome<br>5: Portrait<br>6: Landscape<br>In case of Custom Picture Control, set the base Picture Control. |
| MonochromeFlag | 1 | Monochrome Flag<br>0: color<br>1: monochrome |
| CustomFlag | 1 | Custom Flag<br>0 : Standard<br>1 : Custom<br>2 : Unused custom |
| RegistrationName | 20 | Registration name of Picture Control<br>The string data is 20 byte fixation, and null terminated.<br>(20 characters in actual.) |
| QuickAdjustFlag | 1 | Quick Adjust Flag<br>0: invalid<br>1: valid<br>In case of ulPicCtrlItem of NkMAIDPicCtrlData is Neutral or Custom Picture Control, it is 0 fixation. |
| QuickAdjust | 1 | Quick Adjust value（1step）<br>-2 to +2 |
| Saturation | 1 | Saturation（0.25step）<br>-3 to +3<br>-128 is Auto |
| Hue | 1 | Hue（0.25step）<br>-3 to +3 |
| Sharpening | 1 | Sharpening（0.25step）<br>0 to 9 |

| | | |
|---|---|---|
| | | -128 is Auto |
| Contrast | 1 | Contrast（0.25step）<br>-3 to +3<br>-128 is Auto |
| Brightness | 1 | Brightness（0.25step）<br>-1.5 to +1.5 |
| Clarity | 1 | Clarity （0.25step）<br>-5 ～ +5<br>-128：Auto |
| CustomCurveFlag | 1 | Custom Curve Flag（0.25step）<br>0: No Custom Curve<br>1: Custom Curve used |
| CustomCurveData | 578 | Custom Curve Data<br>This data is not added when there is no Custom Curb.<br>[Header] 64 byte + [LUT] 257x 2 byte = 578 byte<br>Refer to "LUT format" for details. |

## [Monochrome] Second generation

| Field | Size (Byte) | Data |
|---|---|---|
| PicCtrlItem | 1 | type of Picture Control<br>1: Standard<br>2: Neutral<br>3: Vivid<br>4: Monochrome<br>5: Portrait<br>6: Landscape<br>In case of Custom Picture Control, set the base Picture Control. |
| MonochromeFlag | 1 | Monochrome Flag<br>0: color<br>1: monochrome |
| CustomFlag | 1 | Custom Flag<br>0 : Standard<br>1 : Custom<br>2 : Unused custom |
| RegistrationName | 20 | Registration name of Picture Control<br>The string data is 20 byte fixation, and null terminated. |

| | | |
|---|---|---|
| | | (19 characters in actual.) |
| FilterEffects | 1 | Filter Effect<br>0: None<br>1: Yellow<br>2: Orange<br>3: Red<br>4: Green |
| Toning | 1 | Toning(ToneColor)<br>0:B&W<br>1:Sepia<br>2:Cyanotype<br>3:Red<br>4:Yellow<br>5:Green<br>6:Blue Green<br>7:Blue<br>8:Purple Blue<br>9:Red Purple |
| ToningDensity | 1 | Toning(Level)(0.25step)<br>1 to 7<br>When Toning is B&W, this is not referd. |
| Reserve | 1 | Vacant |
| Sharpening | 1 | Sharpening（0.25step）<br>0 to 9<br>-128 is Auto |
| Contrast | 1 | Contrast（0.25step）<br>-3 to +3<br>-128 is Auto |
| Brightness | 1 | Brightness（0.25step）<br>-1.5 to +1.5 |
| Clarity | 1 | Clarity （0.25step）<br>-5 ～ +5<br>-128：Auto |
| CustomCurveFlag | 1 | Custom Curve Flag（0.25step）<br>0：No Custom Curve<br>1：Custom Curve used |
| CustomCurveData | 578 | Custom Curve Data |

| | | |
|---|---|---|
| | | This data is not added when there is no Custom Curb.<br>[Header] 64 byte + [LUT] 257x 2 byte = 578 byte<br>Refer to "LUT format" for details. |

I will show below the value to set the format of the picture control data and the value that identifies the camera.

The street below for value settings for each step width.

[In case of step width of 0.25]

| Picture Control Format | Value that identifies the camera |
|---|---|
| -20 | -5 |
| -19 | -4.75 |
| -18 | -4.5 |
| -17 | -4.25 |
| -16 | -4 |
| -15 | -3.75 |
| -14 | -3.5 |
| -13 | -3.25 |
| -12 | -3 |
| -11 | -2.75 |
| -10 | -2.5 |
| -9 | -2.25 |
| -8 | -2 |
| -7 | -1.75 |
| -6 | -1.5 |
| -5 | -1.25 |
| -4 | -1 |
| -3 | -0.75 |
| -2 | -0.5 |
| -1 | -0.25 |
| 0 | 0 |
| 1 | 0.25 |
| 2 | 0.5 |
| 3 | 0.75 |
| 4 | 1 |
| 5 | 1.25 |
| 6 | 1.5 |
| 7 | 1.75 |
| 8 | 2 |
| 9 | 2.25 |
| 10 | 2.5 |
| 11 | 2.75 |
| 12 | 3 |
| 13 | 3.25 |
| 14 | 3.5 |
| 15 | 3.75 |
| 16 | 4 |
| 17 | 4.25 |
| 18 | 4.5 |
| 19 | 4.75 |
| 20 | 5 |
| 21 | 5.25 |

| | |
|---|---|
| 22 | 5.5 |
| 23 | 5.75 |
| 24 | 6 |
| 25 | 6.25 |
| 26 | 6.5 |
| 27 | 6.75 |
| 28 | 7 |
| 29 | 7.25 |
| 30 | 7.5 |
| 31 | 7.75 |
| 32 | 8 |
| 33 | 8.25 |
| 34 | 8.5 |
| 35 | 8.75 |
| 36 | 9 |

[In case of step width of 1]

| Picture Control Format | Value that identifies the camera |
|---|---|
| -3 | -3 |
| -2 | -2 |
| -1 | -1 |
| 0 | 0 |
| 1 | 1 |
| 2 | 2 |
| 3 | 3 |
| 4 | 4 |
| 5 | 5 |
| 6 | 6 |
| 7 | 7 |
| 8 | 8 |
| 9 | 9 |

## [LUT format]

LUT data is composed from LUT and LUT header. LUT is 2048 byte 11 bit * 8 bit, LUT header is 64 byte and is used by host. LUT header format is original specification by host, and the camera is not concerned of the content of LUT header. But, the top 2 byte of LUT header is used for camera to judge LUT header exist or not. So, the client has to set LUT header.

| Byte | Contents |
|---|---|
| 0, 1 | Length(2116) |
| 2, 3 | Reserved |
| 4 ～ 67 | Lut Header |
| 68 | Data0 |
| 69 | Data1 |
| ・・・ | |
| 2115 | Data2047 |

## [LUT header format]

The content of the LUT header set by the application made of Nikon is shown below as the example.

| Byte | contents | Range |
|---|---|---|
| 1 | AriaID (Byte1) | 0x49 |
| 2 | AriaID (Byte2) | 0x30 |
| 3 | Input Minimum (Black Point) | 0-255 |
| 4 | Input Maximum | 0-255 |
| 5 | Output Minimum | 0-255 |
| 6 | Output Maximum | 0-255 |
| 7 | Gamma (integer portion) | 0-20 |
| 8 | Gamma (fractional portion) | 0-100 |
| 9 | Number of Spline Points | 2-20 |
| 10、11 | Splime Point1 (x、y) | 0-255、0-255 |
| 12、13 | Splime Point2 (x、y) | 0-255、0-255 |
| ・・・ | | |
| 48、49 | Splime Point20 (x、y) | 0-255、0-255 |
| 50 ～ 64 | Reserved | 0 |

### 3.26. PictureControlDataEx

This will get or set the assigned Picture Control data of the second generation.
(Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_PictureControlDataEx |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Generic |
| **ulOperations** | kNkMAIDCapOperation_Set, kNkMAIDCapOperation_Get<br>kNkMAIDCapOperation_GetDefault |
| **Data** | pointer to NkMAIDPicCtrlData structure<br>typedef struct tagNkMAIDPicCtrlData<br>{<br>ULONG ulPicCtrlItem;------ The target Picture Control<br>ULONG ulSize;----------The size of Picture Control data (Max: 610 bytes)<br>bool bModifiedFlag; ----------Modification flag(false: initial registration, true: edit)<br>void* pData;---------- The pointer of Picture Control data.<br>} NkMAIDPicCtrlData, FAR* LPNkMAIDPicCtrlData; |

The range of value sets to “ulPicCtrlItem” is enumerated by Capability_PictureControl.

The following cases, this Capability is read only.

- During movie file recording.

The following cases, refer to Capability_PictureContorolData.

- In case of set
- How to use in case of get
- The format of picture control
- The format of LUT

### 3.27. GetPicCtrlInfo

This will get the Picture Control information. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_GetPicCtrlInfo |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Generic |
| **ulOperations** | kNkMAIDCapOperation_Get |
| **Data** | pointer to NkMAIDGetPicCtrlInfo structure<br>typedef struct tagNkMAIDGetPicCtrlInfo<br>{<br>ULONG ulPicCtrlItem;------The target Picture Control<br>ULONG ulSize;-----The size of Picture Control information (39bytes fixation)<br>void* pData;---------The pointer of Picture Control information.<br>} NkMAIDGetPicCtrlInfo, FAR* LPNkMAIDGetPicCtrlInfo; |

The client must set the all the member of NkMAIDGetPicCtrlInfo.

The value range of Picture Control set to “ulPicCtrlItem” is enumerated by Capability_ PictureControl.

The Picture Control information is valid when “ulPicCtrlItem” is color.

If “ulPicCtrlItem” is monochrome or there is no picture control of base, the Picture Control information will be all 0.

The format of the Picture Control information is shown below.

## [Picture Control information]

| Offset | Size | Field | Data | Description | |
|---|---|---|---|---|---|
| 0x00 | 1 | ValidFlag | 0 : invalid<br>1 : valid | It shows whether the data valid or invalid.<br>When there is no base Picture Control or<br>when it is monochrome, this value is 0. | |
| 0x01 | 1 | QuickCapa | 0x80 : selectable<br>0x01 : AUTO usable<br>0x81 : selectable & AUTO usable | Quick Adjust setting. | |
| 0x02 | 1 | SharpenessCapa | 0x80 : selectable<br>0x01 : AUTO usable<br>0x81 : selectable & AUTO usable | Sharpening setting | |
| 0x03 | 1 | ClarityCapa | 0x80 : selectable<br>0x01 : AUTO usable<br>0x81 : selectable & AUTO usable | Articulation setting | |
| 0x04 | 1 | ContrastCapa | 0x80 : selectable<br>0x01 : AUTO usable<br>0x81 : selectable & AUTO usable | Contrast setting | |
| 0x05 | 1 | BrightnessCapa | 0x80 : selectable<br>0x01 : AUTO usable<br>0x81 : selectable & AUTO usable | Brightness setting | |
| 0x06 | 1 | SaturationCapa | 0x80 : selectable<br>0x01 : AUTO usable<br>0x81 : selectable & AUTO usable | Saturation setting | |
| 0x07 | 1 | HueCapa | 0x80 : selectable<br>0x01 : AUTO usable<br>0x81 : selectable & AUTO usable | Hue setting | |
| 0x08 | 1 | DefaultQuickLevel | -2～+2 | Quick Adjust default value | |
| 0x09 | 1 | DefaultLevel[0] | 0～9 | Quick Adjust | Sharpening |
| 0x0A | 1 | | -5～+5 | | Aarticulation |
| 0x0B | 1 | | -3～+3 | | Contrast |
| 0x0C | 1 | | -1.5～+1.5 | | Brightness |
| 0x0D | 1 | | -3～+3 | | Saturation |

| | | | | | |
|---|---|---|---|---|---|
| 0x0E | 1 | | -3～+3 | -2 | Hue |
| 0x0F | 1 | DefaultLevel[1] | 0～9 | Quick Adjust | Sharpening |
| 0x10 | 1 | | -5～+5 | | Aarticulation |
| 0x11 | 1 | | -3～+3 | | Contrast |
| 0x12 | 1 | | -1.5～+1.5 | | Brightness |
| 0x13 | 1 | | -3～+3 | | Saturation |
| 0x14 | 1 | | -3～+3 | -1 | Hue |
| 0x15 | 1 | DefaultLevel[2] | 0～+9 | Quick Adjust | Sharpening |
| 0x16 | 1 | | -5～+5 | | Aarticulation |
| 0x17 | 1 | | -3～+3 | | Contrast |
| 0x18 | 1 | | -1.5～+1.5 | | Brightness |
| 0x19 | 1 | | -3～+3 | | Saturation |
| 0x1A | 1 | | -3～+3 | 0 | Hue |
| 0x1B | 1 | DefaultLevel[3] | 0～9 | Quick Adjust | Sharpening |
| 0x1C | 1 | | -5～+5 | | Aarticulation |
| 0x1D | 1 | | -3～+3 | | Contrast |
| 0x1E | 1 | | -1.5～+1.5 | | Brightness |
| 0x1F | 1 | | -3～+3 | | Saturation |
| 0x20 | 1 | | -3～+3 | 1 | Hue |
| 0x21 | 1 | DefaultLevel[4] | 0～+9 | Quick Adjust | Sharpening |
| 0x22 | 1 | | -5～+5 | | Aarticulation |
| 0x23 | 1 | | -3～+3 | | Contrast |
| 0x24 | 1 | | -1.5～+1.5 | | Brightness |
| 0x25 | 1 | | -3～+3 | | Saturation |
| 0x26 | 1 | | -3～+3 | 2 | Hue |

### 3.28. DeleteCustomPictureControl

This will delete Custom Picture Control. (Shooting menu)

**Capability** kNkMAIDCapability_DeleteCustomPictureControl

**Object types** Source

**ulType** kNkMAIDCapType_Unsigned

**ulOperations** kNkMAIDCapOperation_Set

**Data** Custom Picture Control Item

When the client set the one of Custom Picture Control enumerated by Capability_PictureControl and executes kNkMAIDCapOperation_Set, the specified Custom Picture Control will be deleted.

This Capability can not set during movie file recording.

### 3.29. Active_D_Lighting

This will set Active D-Lighting. (Shooting menu)

**Capability** kNkMAIDCapability_Active_D_Lighting

**Object types** Source

**ulType** kNkMAIDCapType_Unsigned

**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault

**Data** one of meNkMAIDActive_D_Lighting

| |
|---|
| 0: High |
| 1: Normal |
| 2: Low |
| 3: Off |
| 4: Extra high |
| 5: Auto |

The following cases, this Capability can not set.

- When the Capability_ExposureMode is Scene Modes or Special Effects Modes.
- When the Capability_InfoDisplayErrStatus is True: ON (Error display).
- During movie file recording.

### 3.30. ISOAutoShutterTime

This will set the shutter speed when ISO is controlled automatically. (Shooting menu)

**Capability** kNkMAIDCapability_ISOAutoShutterTime

**Object types** Source

**ulType** kNkMAIDCapType_Unsigned

**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set

**Data** one of eNkMAIDISOAutoShutterTime

| Shutter speed | eNkMAIDISOAutoShutterTime |
|---|---|
| 1/2000 | 23 |
| 1/1600 | 24 |
| 1/1250 | 25 |
| 1/1000 | 26 |
| 1/800 | 27 |
| 1/640 | 28 |
| 1/500 | 29 |
| 1/400 | 30 |
| 1/320 | 31 |
| 1/250 | 13 |
| 1/200 | 14 |
| 1/160 | 15 |
| 1/125 | 0 |
| 1/100 | 16 |
| 1/80 | 17 |
| 1/60 | 1 |
| 1/50 | 19 |
| 1/40 | 18 |
| 1/30 | 2 |
| 1/15 | 3 |
| 1/8 | 4 |
| 1/4 | 5 |
| 1/2 | 6 |
| 1 | 7 |
| 2 | 8 |

| | |
|---|---|
| 4 | 9 |
| 8 | 10 |
| 15 | 11 |
| 30 | 12 |
| Auto | 32 |

The following cases, this Capability cannot be set into.

- When the Capability_ExposureMode is Scene Modes or Special Effects Modes.
- During movie file recording.
- When the Capability_IsoControl is False.

### 3.31. ISOAutoShutterTimeAutoValue

This will set compensation value when the value of kNkMAIDCapability_ISOAutoShutterTime is “Auto”. (Shooting menu)

**Capability** kNkMAIDCapability_ISOAutoShutterTimeAutoValue
**Object types** Source
**ulType** kNkMAIDCapType_Range
**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
**Data** -2～+2EV(Default value: 0)

The following cases, this Capability is read only.

- When the Capability_IsoControl is False.
- When the value of Capability_ISOAutoShutterTime is except “Auto”.
- During movie file recording.
- When the Capability_ExposureMode is Scene Modes or Special Effects Modes.

### 3.32. ISOAutoHiLimit

This will set the max sensitivity when ISO is controlled automatically. (Shooting menu)

**Capability** kNkMAIDCapability_ISOAutoHiLimit

**Object types** Source

**ulType** kNkMAIDCapType_Unsigned

**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set

**Data** one of eNkMAIDISOAutoHiLimit

| eNkMAIDISOAutoHiLimit | ISO |
|---|---|
| 0 | 200 |
| 1 | 400 |
| 2 | 800 |
| 3 | 1600 |
| 4 | 3200 |
| 5 | 6400 |
| 8 | 12800 |
| 9 | 25600 |

The following cases, this Capability cannot be set into.

- When the Capability_ExposureMode is Scene Modes or Special Effects Modes.
- During movie file recording.
- When the Capability_IsoControl is False.

### 3.33. MovieScreenSize

This will set the value “Frame size” of “Movie settings”. (Shooting menu)

**Capability** kNkMAIDCapability_MovieScreenSize

**Object types** Source

**ulType** kNkMAIDCapType_Unsigned

**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set

**Data** one of eNkMAIDMovieScreenSize3

[In the case of D5500]

| eNkMAIDMovieScreenSize3 | Content | |
|---|---|---|
| | NTSC | PAL |
| 0 | 1920×1080 60p | 1920×1080 50p |
| 1 | 1920×1080 30p | 1920×1080 25p |
| 2 | 1920×1080 24p | 1920×1080 24p |
| 3 | 1280× 720 60p | 1280× 720 50p |
| 4 | 640× 424 30p | 640× 424 25p |

[In the case of D5600]

| eNkMAIDMovieScreenSize3 | Content |
|---|---|
| 0 | 1920×1080 60p |
| 1 | 1920×1080 50p |
| 2 | 1920×1080 30p |
| 5 | 1920×1080 25p |
| 6 | 1920×1080 24p |
| 3 | 1280×720 60p |
| 7 | 1280×720 50p |

When during movie file recording, this Capability is cannot be set into.

### 3.34. MovieRecMicrophone

This will set the shooting menu, [Movie settings – Microphone]. (Shooting menu)

**Capability** kNkMAIDCapability_MovieRecMicrophone

**Object types** Source

**ulType** kNkMAIDCapType_Unsigned

**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set

**Data** one of eNkMAIDMovRecMicrophone

0: Auto sensitivity

4: Microphone off

5: Manual sensitivity

The following cases, this Capability is read only.

- During movie file recording.

### 3.35. MovieRecMicrophoneValue

This will set the microphone sensitivity when the Capability_MovieRecMicrophone is manual sensitivity. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_MovieRecMicrophoneValue |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Range |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set |
| **Data** | 1 to 20 step=1 (Default: 15) |

The following cases, this Capability cannot be set into.

- When the value of Capability_MovieRecMicrophone is not 5 (Manual sensitivity)
- During movie file recording.

### 3.36. MovieWindNoiseReduction

This will set the shooting menu, [Movie setting – Wind noise reduction]. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_MovieWindNoiseReduction |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set |
| **Data** | one of eNkMAIDMovieWindNoiseReduction<br>0: OFF<br>1: ON |

The following cases, this Capability cannot be set into.

- During movie file recording.
- When having connected the external microphone.

### 3.37. MovieManualSetting

This will set the shooting menu, [Movie setting – Manual movie settings]. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_MovieManualSetting |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set |
| **Data** | one of eNkMAIDMovManualSetting<br>0 : OFF<br>1 : ON |

When during movie file recording, this Capability cannot be set into.

### 3.38. MovieImageQuality

This will set the shooting menu, [Movie settings – Movie quality]. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_MovieImageQuality |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set |
| **Data** | one of eNkMAIDMovieImageQuality<br>0: Normal<br>1: High quality |

When during movie file recording, this Capability cannot be set into.

### 3.39. AutoDistortion

This will set the shooting menu, [Auto distortion control]. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_AutoDistortion |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set |
| **Data** | one of eNkMAIDAutoDistortion<br>0: Off<br>1: On |

The following cases, this Capability cannot be set into.

- During movie file recording.
- When the CPU lens does not attached.
- When the lens unsupported auto distortion control is attached.

### 3.40. HDRMode

This will set the HDR(high dynamic range). (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_HDRMode |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set |
| **Data** | one of eNkMAIDHDRMode<br>0: Off<br>1: Low<br>2: Normal<br>3: High<br>4: Extra high<br>5: Auto |

The following cases, this Capability cannot be set into.

- When the Capability_ExposureMode is Scene Modes or Special Effects Modes.
- When the value of Capability_CompressionLevel is “RAW” or “RAW+ JPEG (Basic/Normal/Fine)”.
- When the Capability_EnableBracketing is True (ON).
- When the Capability_InfoDisplayErrStatus is True: ON (Error display).
- During movie file recording.

### 3.41. SceneMode

This will set the shooting menu, [Scene mode]. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_SceneMode |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set |
| **Data** | one of eNkMAIDSceneMode<br>0: Night Landscape<br>1: Party/Indoor<br>2: Beach/Snow<br>3: Sunset<br>4: Dusk/Dawn<br>5: Pet Portrait<br>6: Candlelight<br>7: Blossom<br>8: Autumn Colors<br>9: Food<br>13: Portrait<br>14: Landscape<br>15: Child<br>16: Sports<br>17: Close up<br>18: Night portrait |

The value of this capability selects Scene Modes, when the mode dial is rotated to "SCENE".

The value of Capability_ExposureMode and the value of this capability are the same "Scene Mode".

The Follwing cases, this Capability cannot be set into.

- When the Capability_ExposureMode is not "14: SCENE".
- During movie file recording.

### 3.42. EffectMode

This will get the shooting menu, [Special Effects]. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_EffectMode |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set |
| **Data** | one of eNkMAIDEffectMode<br>0: Night Vision<br>2: Miniature Effect<br>3: Selective Color<br>4: Silhouette<br>5: High Key<br>6: Low Key<br>7: Toy Camera Effect<br>9: Super Vivid<br>10: Pop<br>11: Photo Illustration |

The following cases, this Capability cannot be set into.

- During movie file recording.
- When the Capability_ExposureMode is not “17: EFFECTS”.

### 3.43. VignetteControl

This will set the Vignette control. (Shooting menu)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_VignetteControl |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set |
| **Data** | one of eNkMAIDVignetteControl<br>0: High<br>1: Normal<br>2: Low<br>3: Off |

The following cases, this Capability cannot be set into.

- During movie file recording.

### 3.44. AFcPriority

This will get the custom settings menu, [Autofocus – AF-C priority selection].

**Capability** kNkMAIDCapability_AFcPriority

**Object types** Source

**ulType** kNkMAIDCapType_Enum
kNkMAIDArrayType_PackedString

**ulOperations** kNkMAIDCapOperation_Get,
kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set

**Data**

| | |
|---|---|
| Focus | "AF-C Focus" |
| Release | "AF-C Shutter" |

When the Capability_LiveViewStatus is 1(ON), this capability is set to read-only.

### 3.45. AFAreaPoint

This will get the custom settings menu, [Autofocus –Number of focus points].

**Capability** kNkMAIDCapability_AFAreaPoint

**Object types** Source

**ulType** kNkMAIDCapType_Unsigned

**ulOperations** kNkMAIDCapOperation_Get,
kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault

**Data** one of eNkMAIDAFAreaPoint
1: 11 points
2: 39 points

When Capability_LiveViewStatus is 1(On), this capability is set to read-only.

### 3.46. EVInterval

This will set the EV steps for exposure control.

**Capability** kNkMAIDCapability_EVInterval

**Object types** Source

**ulType** kNkMAIDCapType_Enum
kNkMAIDArrayType_PackedString

**ulOperations** kNkMAIDCapOperation_Get,
kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set

**Data**

| | |
|---|---|
| 1/3 step | "1/3 Step" |
| 1/2 step | "1/2 Step" |

When this capability is changed and Capability_BracketingVary is set to AE bracketing, Capability_AEBracketingStep will be set to 1/ EV (3), and Capability_EnableBracketing is set to OFF (False).

The following cases, this Capability cannot be set into.

- During movie file recording.

### 3.47. BracketingVary

This will select the bracketing variation.

**Capability** kNkMAIDCapability_BracketingVary

**Object types** Source

**ulType** kNkMAIDCapType_Enum
kNkMAIDArrayType_PackedString

**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set

**Data**

| | |
|---|---|
| AE bracketing | "AE Only" |
| WB bracketing | "White Balance" |
| ADL bracketing | "ADL bracketing" |

The following cases, this Capability cannot be set into.

- When the Capability_ExposureMode is Scene Modes or Special Effects Modes.
- If the Capability_HDRMode is set to other than "0: Off".
- When the Capability_InfoDisplayErrStatus is True: ON (Error display).

・ During movie recording.

### 3.48. InternalSplMode

This will show “Flash, Bracketing, Flash cntrl for built-in flash/Optional flash” of Custom Settings.

**Capability** kNkMAIDCapability_InternalSplMode

**Object types** Source

**ulType** kNkMAIDCapType_Enum
kNkMAIDArrayType_PackedString

**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set

**Data**

| | |
|---|---|
| TTL mode | “TTL” |
| Manual luminescence mode | “Manual” |
| Command mode | “Command” |

When the value of Capability_ExternalFlashSort is “3: Not exist” or “4: new communication (without setting display:SB-400)”, if you have installed an external flash commander mode not supported and if new communication (no operation setting display member), the two items of "TTL", "Manual" The value of this property is limited.

The following cases, this Capability cannot be set into.

・ When the Capability_ExposureMode is EFFECTS.
・ During movie recording.

### 3.49. VideoMode (Only D5500 is supported)

This will set the Video mode.(SETUP)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_VideoMode |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault |
| **Data** | one of eNkMAIDVideoMode<br>0: NTSC<br>1: PAL |

When during movie file recording, this Capability cannot be set into.

### 3.50. UserComment

This will set a description of an image. (SETUP)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_UserComment |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_String |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set |
| **Data** | String shorter than 36 bytes (not including termination '¥0') |

If the client set string longer than 36 bytes, the module uses 36 bytes from the head.

The character, which can be included in the string, is only an ASCII characters in the following table. When the other character is set, the module returns an error(kNkMAIDResult_ValueOutOfBounds).

When during movie file recording, this Capability cannot be set into.

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SP | ! | " | # | $ | % | & | ' | ( | ) | * | + | , | - | . | / |
| : | ; | < | = | > | ? | @ | [ | ] | _ | { | } | | | | |
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | | | | | | |
| A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P |
| Q | R | S | T | U | V | W | X | Y | Z | | | | | | |
| a | b | C | d | e | f | g | h | i | j | k | l | m | n | o | p |
| q | r | S | t | u | v | w | x | y | z | | | | | | |

### 3.51. EnableComment

This will enable to add UserComment to an image file. (SETUP)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_EnableComment |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Boolean |
| **ulOperations** | kNkMAIDCapOperation_Get,<br>kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault |
| **Data** | True: Enable  False: Disable |

When during movie file recording, this Capability cannot be set into.

### 3.52. EnableCopyright

This will set whether attach copyright information. (SETUP)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_EnableCopyright |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Boolean |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set,<br>kNkMAIDCapOperation_GetDefault |
| **Data** | True: attach  False: none |

The following cases, this Capability cannot be set into.

- During movie file recording.

### 3.53. ArtistName

This will set the artist information. (SETUP)

**Capability** kNkMAIDCapability_ArtistName
**Object types** Source
**ulType** kNkMAIDCapType_String
**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
**Data** String shorter than 36 bytes (not including termination '/0')

If the client set string longer than 36 bytes, the module uses 36 bytes from the head, and the space (0x20) before termination '¥0' of string will be ignored.
for example.)

ABCD□EFG□□□□□□□□'¥0'

□ shows a space (0x20) , so 8 space is ignored.

The character, which can be included in the string, is only an ASCII 90 characters. (refer to the table in the ShootingBankName.) When the other character is set, the module returns an error (kNkMAIDResult_ValueOutOfBounds).

The following cases, this Capability cannot be set into.

- During movie file recording.

### 3.54. CopyrightInfo

This will set the copyright information. (SETUP)

**Capability** kNkMAIDCapability_CopyrightInfo
**Object types** Source
**ulType** kNkMAIDCapType_String
**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set
**Data** String shorter than 54 bytes (not including termination '/0')

If the client set string longer than 54 bytes, the module uses 55 bytes from the head, and the space(0x20) before termination '¥0' of string will be ignored.
for example.)

ABCD□EFG□□□□□□□□'¥0'

□ shows a space (0x20), so 8 spaces are ignored.

The character, which can be included in the string, is only an ASCII 90 characters. (refer to the table in the ShootingBankName.) When the other character is set, the module returns an error (kNkMAIDResult_ValueOutOfBounds).

The following cases, this Capability cannot be set into.

- During movie file recording.

### 3.55. CameraInclinationMode

This will set whether add or not rotate information to the image file.
(D5500 : SETUP, D5600 : PLAYBACK)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_CameraInclinationMode |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Boolean |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault |
| **Data** | True: Add False: not Add |

When the value of this capability is set to False, the Capability_CameraInclination is always zero (Level).

When during movie file recording, this Capability cannot be set into.

### 3.56. ClockDateTime

This will set the built-in clock of camera. (SETUP)

| | |
|---|---|
| **Capability** | kNkMAIDCapability_ClockDateTime |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_DateTime |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set |
| **Data** | kNkMAIDDataType_DateTimePtr<br>[D5500]20150101T000000: 2015/01/01 00:00:00<br>[D5600]20160101T000000: 2016/01/01 00:00:00 |

When during movie file recording, this Capability cannot be set into.

### 3.57. ShutterSpeed

This will set the shutter speed.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_ShutterSpeed |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Enum |
| | kNkMAIDArrayType_PackedString |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, |
| | kNkMAIDCapOperation_Set |
| **Data** | Strings of shutter time in second. (e.g.) ”1”, “1/1.3”, “1/1.6”, |
| | “x 1/250”, “x 1/200”, |

The information whether the exposure is over or under can not be acquired by getting only the value of this capability. In the condition that the information of the shutter speed being blinking is acquired form the value of Capability_BlinkingStatus, the exposure is over if the Capability_ShutterSpeed is maximum value. The exposure is under in the same condition if the Capability_ShutterSpeed is minimum value instead.

When the Capability_MovieManualSetting is "1 ON", the value of ExposureMode is “Manual” and running LiveView, the range of values should be changed as follows.

| Framerate | Shutter speed |
|---|---|
| 24p、25p、30p | 1/4000～1/30 |
| 50p、50i | 1/4000～1/50 |
| 60p、60i | 1/4000～1/60 |

In any of the following, the “Bulb, Time” cannot be set into.

- When the Capability_ExposureMode is not “Manual”.
- When the Capability_HDRMode is not 0 (Off)

The following cases, this Capability cannot be set into.

- When the Capability_ExposureMode is “Program”, “Aperture Priority”, “Scene Mode” or “EFFECTS”.
- When sequence error has occurred.
- When the Capability_RetractableLensWarningStatus is “True: Warning about retraction”.

### 3.58. FlexibleProgram

This will set the Flexible program value.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_FlexibleProgram |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Range |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set |
| **Data** | -5～+5EV（Default value: 0） |

The module set the step values same as Capability_EVInterval. When the Capability_EVInterval, Capability_ExposureMode is changed, the capability is set to default(0), and the module sends to the client kMAIDEvent_CapChange or kMAIDEvent_CapChangeValueOnly.

The following cases, ulVisibility of this capability is invalid and this capability is set to read-only.

- When the Capability_ExposureMode is not “Program”.
- When sequence error has occurred.
- When the Capability_RetractableLensWarningStatus is true.

### 3.59. FocusPreferredArea

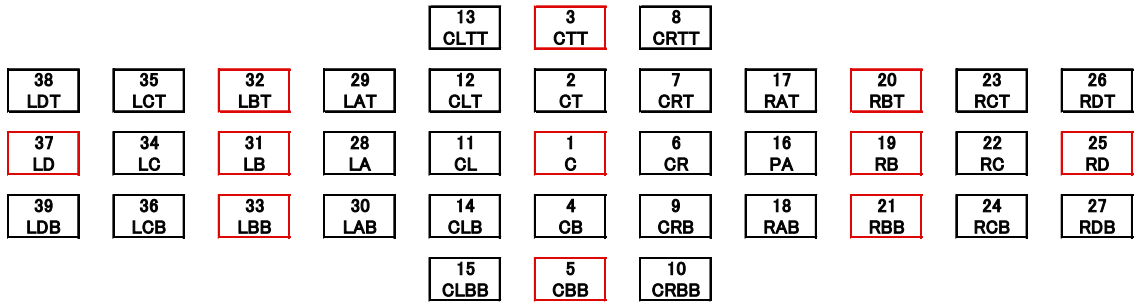
This will select the preferred focus area.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_FocusPreferredArea |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Enum |
| | kNkMAIDArrayType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, |
| | kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set |
| **Data** | one of eNkMAIDFocusPreferred4 |
| | 0 – 39 (default 1) |

When the value of this capability is 0, it is shown that the focus point is not decided.

The relationship between focus point and the value of this capability is shown in following figure.

When the Capability_AFAreaPoint is 11 points, the range is 0-11 can be selected by this Capability.




The following cases, this Capability cannot be set into.

- When the Capability_FocusAreaMode is “Auto-area AF”.
- When the value of Capability_LiveViewStatus is 1(ON).

### 3.60. Aperture

This will set the aperture.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_Aperture |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_EnumkNkMAIDArrayType_PackedString |
| **ulOperations** | kNkMAIDCapOperation_Get,<br>kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set |
| **Data** | String of F value (e.g.) “1.4”, “1.6”, “1.8”... |

When aperture is not set to minimum(FEE), this capability is read-only and the string of “FEE” is set. When this capability is “FEE”, the module can’t execute capture-command.

When sequence error has occurred, the ulVisibility of this capability is set to invalid and ulOperations of this capability is set to read-only and the current value is invalid. If the ulOperations is changed, the module sends to the client kMAIDEvent_CapChange.

The information whether the exposure may be over or under can not be acquired by getting only the value of this capability. In the condition that the information of the aperture being blinking is acquired form the value of Capability_BlinkingStatus, the exposure is over if the Capability_ Aperture is minimum value. The exposure is under in the same condition if the Capability_ShutterSpeed is maximum value instead.

In one case of following, this capability cannot be set into.

- In Program, Speed Priority, Scene Mode, or Special Effects Modes, the Capability_ExposureMode is.
- When the lens is not set as the minimum iris diaphragm. (The state of the lens is “FEE”.)
- When a sequence error has occurred
- In the lens case of not equipping.
- When Capability_RetractableLensWarningStatus is “True: Warning about retraction”.

### 3.61. MeteringMode

This will get the metering mode.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_MeteringMode |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get,<br>kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault |
| **Data** | one of eNkMAIDMeteringMode<br>0: Matrix<br>1: Center weighted<br>2: Spot |

When the CPU lens is not attached, 1(Center weighted) will be applied for metering mode.

The following cases, this Capability is set to read-only.

- When the Capability_ExposureMode is Scene Modes or Special Effects Modes.
- When AE locked.
- During movie file recording.
- When the value of Capability_LiveViewImageZoomRate is not “0: Whole display”.

### 3.62. ExposureMode

This will select the exposure mode.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_ExposureMode |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_EnumkNkMAIDArrayType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get,<br>kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set |
| **Data** | one of eNkMAIDExposureMode<br>0: Program mode<br>1: Aperture priority<br>2: Speed priority<br>3: Manual<br>5: [Scene Modes]Auto<br>13: [Scene Modes] Auto(flash off)<br>14: [Scene Modes]SCENE Other scenes<br>17: [Special Effects]EFFECTS |

The value of 5, 13 and 14 is called Scene Modes. If [14: SCENE Other scenes] is set, the Scene Modes set by Capability_SceneMode will be used.

If [17: Special Effects] is set, the Special Effects Mode set by Capability_EffectMode will be used.

Shooting mode listed during Live View is composed of “Program mode”, “Aperture priority”, “Speed priority”, and “Manual”.

The following cases, this Capability cannot be set into

- When the Capability_LockCamera is false.
- During movie recording.

### 3.63. ExposureComp

This will set the exposure compensation value.

**Capability** kNkMAIDCapability_ExposureComp

**Object types** Source

**ulType** kNkMAIDCapType_Range

**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

**Data** -5～+5EV（Default value: 0）

The module sets the step value same as Capability_EVInterval. When the Capability_EVInterval is changed, the module sends to the client kMAIDEvent_CapChange.

If any of the following, this Capability cannot be set into.

- When the Capability_ExposureMode is Scene Modes or the value other than "Night Vision" of Special Effects Modes.
- When the Capability_InfoDisplayErrStatus is True (ON).

### 3.64. ShootingMode

This will set the shooting mode.

**Capability** kNkMAIDCapability_ShootingMode

**Object types** Source

**ulType** kNkMAIDCapType_EnumkNkMAIDArrayType_Unsigned

**ulOperations** kNkMAIDCapOperation_Get,
kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set

**Data** one of eNkMAIDShootingMode

| |
|---|
| 0: SingleFrame |
| 1; Continuous L |
| 2: Continuous H |
| 3: Self-timer |
| 5: Quick-response remote (Only D5500 is supported) |
| 6: Delayed remote (Only D5500 is supported) |
| 8: Quiet |

If any of the following, the ulOperations cannot be set into.

- When the Capability_InfoDisplayErrStatus is True: ON (Error display).
- During movie file recording.

### 3.65. ContinuousShootingNum

This will set the number of shots in continuous shooting by host.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_ContinuousShootingNum |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set<br>kNkMAIDCapOperation_GetDefault |
| **Data** | 1<br>(same as the default value of Capability_RemainContinuousShooting) |

The maximum value of this capability is the same as the default value of Capability_RemainContinuousShooting.

When the value of Capability_EnableBracketing is ON and execute bracketing on continuous mode, the client must set the value more than the bracketing number of shot to this capability. But if the client sets the value more than the bracketing number of shot, bracketing will be stop at the setting the bracketing number of shot on continuous mode.

When during movie file recording, this Capability cannot be set into.

### 3.66. FocusAreaMode

This will select the AF area mode for phase detection on still image shooting.

**Capability** kNkMAIDCapability_FocusAreaMode

**Object types** Source

**ulType** kNkMAIDCapType_Enum
kNkMAIDArrayType_PackedString

**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,
kNkMAIDCapOperation_Set

**Data**

| | |
|---|---|
| 9-points dynami-area AF | "Dynamic(9 points)" |
| Single-point AF | "Single" |
| Auto-areaAF | "Auto" |
| 3D – tracking | "3D-tracking" |
| 21-points dynami-area AF | "Dynamic(21 points)" |
| 39-points dynami-area AF | "Dynamic(39 points)" |

In the following table, the default value is changed by Scene modes or Special Effects Modes.

When the setting of Capability_ExposureMode is changed to Scene Modes or Special Effects Modes, the value of this capability will be changed to each default value.

| Capability_ExposureMode<br>Capability_SceneMode<br>Capability_EffectMode | Default |
|---|---|
| Auto(Scene Modes)<br>Auto(flash off) (Scene Modes)<br>Portrait(Scene Modes)<br>Landscape(Scene Modes)<br>Child(Scene Modes)<br>Night Landscape(SCENE)<br>Party/Indoor(SCENE)<br>Beach/Snow(SCENE)<br>Sunset(SCENE)<br>Dusk/Dawn(SCENE)<br>Blossom(SCENE)<br>Autumn Colors(SCENE)<br>Night Portrait(SCENE)<br>Color Sketch(EFFECTS)<br>Selective Color(EFFECTS)<br>Toy Camera (EFFECTS)<br>Super Vivid(EFFECTS)<br>Pop(EFFECTS)<br>Photo Illustration(EFFECTS) | Auto |
| Night Vision(EFFECTS) | Single |

| | |
|---|---|
| Miniature Effect(EFFECTS) | (unchangeable) |
| Close up(Scene Modes)<br>Candlelight(SCENE)<br>Food(SCENE)<br>Silhouette(EFFECTS)<br>High Key(EFFECTS)<br>Low Key(EFFECTS) | Single |
| Sports(Scene Modes)<br>Pet Portrait(SCENE) | Dynamic(39 points) |

When Capability_AFMode is AF-S(0), "3D-tracking" and "Dynamic(9/21/39points)" cannot be set into.

When the value of this capability is "3D-tracking" or "Dynamic(9/21/39points)" and sets the value of Capability_AFMode to AF-S(0), The value of this capability is changed to "Single".

The following cases, this capability cannot be set into.

- When the value of Capability_AFMode is MF (0).
- The CPU lens is not attached.
- When the Capability_ExposureMode is set to Special Effects Modes and the Capability_EffectMode is set to Night Vision or Miniature Effect.
- When the value of Capability_LiveViewStatus is 1(ON).

### 3.67. EnableBracketing

This will set whether bracketing is active or not.

**Capability** kNkMAIDCapability_EnableBracketing

**Object types** Source

**ulType** kNkMAIDCapType_Boolean

**ulOperations** kNkMAIDCapOperation_Get,
kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault

**Data** True: ON False: OFF

When Capability_BracketingVary does not set into WB bracketing or ADL bracketing, and the value of Capability_EVInterval is changed, the value of this capability is changed to False(OFF).

If any of the following, this Capability cannot be set into.

- When the Capability_ExposureMode is Scene Modes or Special Effects Modes.
- When the value of Capability_BracketingVary is “WB bracketing”, and if Capability_CompressionLevel is either ”RAW”, “RAW+JPEG(Basic)”, “RAW+JPEG(Normal)” or “RAW+JPEG(Fine). (The ulVisibility of this capability is set to invalid.)
- When the value of Capability_HDRMode is except 0 (Off).
- During movie file recording.

### 3.68. AEBracketingStep

This will set the exposure increment for AE, SB, AE/SB bracketing.

**Capability** kNkMAIDCapability_AEBracketingStep

**Object types** Source

**ulType** kNkMAIDCapType_Enum
kNkMAIDArrayType_Unsigned

**ulOperations** kNkMAIDCapOperation_Get,
kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set

**Data** one of eNkMAIDAEBracketingStep
0: 1/3EV
1: 1/2EV
2: 2/3EV
3: 1EV
4: 1+1/3EV

5: 1+1/2EV

6: 1+2/3EV

7: 2EV

When the Capability_EnableBracketing is ON(true) and the Capability_BracketingVary is “AE bracketing”, this capability is valid.

If any of the following, the ulOperations cannot be set into.

- The Capability_EnableBracketing is not “ON”.
- The Capability_BracketingVary is not “AE bracketing”.
- The Capability_ExposureMode is Scene Modes or Special Effects Modes.
- During movie file recording.

If the ulVisibility and ulOperations are changed, the module sends to the client kMAIDEvent_CapChange.

| EVInterval | AEBracketingStep |
|---|---|
| 1/3EV | 1/3EV、2/3EV、1EV、<br>1+1/3EV、1+2/3EV、2EV |
| 1/2 EV | 1/2EV、1EV、1+1/2EV、2EV |

### 3.69. WBBracketingStep

This will set the white balance increment for WB bracketing.

**Capability** kNkMAIDCapability_WBBracketingStep

**Object types** Source

**ulType** kNkMAIDCapType_Enum

kNkMAIDArrayType_Unsigned

**ulOperations** kNkMAIDCapOperation_Get,

kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set

**Data** one of eNkMAIDWBBracketingStep

0: 1Step 1: 2Step 2: 3Step

When the Capability_EnableBracketing is ON(true) and the Capability_BracketingVary is “White Balance”, this capability is valid.

If any of the following, the ulOperations cannot be set into.

- The Capability_EnableBracketing is not “ON”.
- The Capability_BracketingVary is not “White Balance”.
- The Capability_ExposureMode is Scene Modes or Special Effects Modes.
- During movie file recording.

### 3.70. Bracketing Type

This will select the combination bracketing shots and direction when AE, White balance bracketing.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_BracketingType |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Enum |
| | kNkMAIDArrayType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, |
| | kNkMAIDCapOperation_GetArray |
| **Data** | one of eNkMAIDBracketingType |
| | 4: Both 3 |

### 3.71. ADLBracketingType

This will select the bracketing shots when ADL bracketing.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_ADLBracketingType |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Enum |
| | kNkMAIDArrayType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, |
| | kNkMAIDCapOperation_GetArray |
| **Data** | one of eNkMAIDADLBracketingType |
| | 0 : 2 shots ( Off - UserSettings) |

### 3.72. LiveViewStatus

This will start or stop Live view and show status of Live view.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_LiveViewStatus |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault |
| **Data** | one of eNkMAIDLiveViewStatus<br>0: OFF<br>1: ON |

When the client start Live view, the client must set the value of this capability to ON(1). And when the client stop Live view, the client must set the value of this capability to OFF(0).

In case of kNkMAIDCapOperation_Get, the value of this capability will show the current status of Live view.

If the client want to get Live view image by Capability_GetLiveViewImage, the client have to set the value of this capability to ON(1) beforehand.

The client have to check this value before closing Source object, and if the value of this capability is ON(1), have to set to OFF(0).

When the Live view is started, the status of camera will be changed to Lock camera internally, but the value of Capability_LockCamera kept the current value.

The client has to check the value of Capability_LiveViewProhibit, and when the value of Capability_LiveViewProhibit is not 0, Live view will not be started.

When the Capability_ExposureMode is Scene Modes or Special Effects Modes, the ulOperations of this capability cannot be set into.

### 3.73. LiveViewProhibit

This will show the status of Live view prohibition.

**Capability** kNkMAIDCapability_LiveViewProhibit

**Object types** Source

**ulType** kNkMAIDCapType_Unsigned

**ulOperations** kNkMAIDCapOperation_Get

**Data** one of eNkMAIDLiveViewProhibit

The live view prohibition is shown by the OR value of the following definition value.

If non-0, the value of this Capability is prohibited Live view.

| value | Conditions prohibited |
|---|---|
| 0x80000000 | Exposure Mode is non-P,S,A,M. |
| 0x01000000 | When Retractable lens is set, the zoom ring does not extend. |
| 0x00200000 | Bulb warning, ShutterSpeed is Time. |
| 0x00100000 | Card not formatted. |
| 0x00080000 | Card error. |
| 0x00040000 | Card protected.(Recording media is “Card” or “Card + SDRAM”) |
| 0x00020000 | High temperature. |
| 0x00008000 | Capture command is executing. |
| 0x00004000 | Recording media is “Card” or “Card + SDRAM” and when no memory card is inserted in the camera, Release locked setting. |
| 0x00000800 | Non-CPU lens is attached, and ExposureMode is not Manual or Aperture priority. |
| 0x00000200 | TTL error |
| 0x00000100 | battery shortage |
| 0x00000020 | Aperture ring is not minimum. |
| 0x00000004 | Sequence error |

### 3.74. LiveViewImageZoomRate

This will set the zoom rate for Live View image.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_LiveViewImageZoomRate |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Enum |
| | kNkMAIDArrayType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, |
| | kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set |
| **Data** | one of eNkMAIDLiveViewImageZoomRate |
| | 0: Whole display |
| | 1: 25 % |
| | 2: 33 % |
| | 3: 50 % |
| | 4: 66% |
| | 5: 100 % |

When the Live view is started, the value of this capability will be set to default value automatically.

The following cases, the ulVisibility of this capability is invalid and this Capability is set to read-only.

- When the value of Capability_LiveViewStatus is not 1(ON).
- During movie file recording.

### 3.75. LiveViewImageSize

This will set size of Live View image.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_LiveViewImageSize |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Enum |
| | kNkMAIDArrayType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, |
| | kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set |
| **Data** | one of eNkMAIDLiveViewImageSize |
| | 1: QVGA |
| | 2: VGA |

When during movie file recording, this Capability cannot be set into.

### 3.76. CameraInclination

This will get inclination of camera.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_CameraInclination |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault |
| **Data** | one of eNkMAIDCameraInclination |
| | 0: Level (included when the inclination cannot be detected) |
| | 1: Grip is top |
| | 2: Grip is bottom |
| | 3: Level (Up Down) |

When the Capability_CameraInclinationMode is false, or the camera cannot detect inclination of itself, the value of this capability is zero(Level).

### 3.77. RemainContinuousShooting

This will get the number of shot that can be recorded on SDRAM or the card in the continuous shooting mode by the command.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_RemainContinuousShooting |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault |
| **Data** | 0 – 100 (Default: 100) |

The value of this capability is always under the value of Capability_ShootingLimit.

The value of this capability will be changed by the following setting.

- Capability_ImageSize
- Capability_CompressionLevel
- Capability_HDRMode
- Capability_AutoDistortion
- Capability_NoiseReduction
- Capability_SceneMode

### 3.78. RemainCountInMedia

This will get the number of shot that can be saved in Card in current image quality.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_RemainCountInMedia |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault |
| **Data** | 0 — 65535 (Default:0) |

When a card is not inserted, the value of this capability is 0.

The value of this capability is changed by the setting of camera.

### 3.79. LockExposure

This will get lock status of auto exposure.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_LockExposure |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Boolean |
| **ulOperations** | kNkMAIDCapOperation_Get |
| **Data** | True: Lock　　False: Unlock |

### 3.80. LockFocus

This will get lock status of auto focus.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_LockFocus |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Boolean |
| **ulOperations** | kNkMAIDCapOperation_Get |
| **Data** | True: Lock　　False: Unlock |

### 3.81. ExposureStatus

This will get the exposure indicator status of Camera.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_ExposureStatus |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Float |
| **ulOperations** | kNkMAIDCapOperation_Get |
| **Data** | ExposureValue (EV)　step = 1/12 (EV) |

### 3.82. InfoDisplayErrStatus

This will show error display status on the information panel.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_InfoDisplayErrStatus |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Boolean |
| **ulOperations** | kNkMAIDCapOperation_Get |
| **Data** | True: ON (Error display) False: OFF |

The value of this capability is updated only if the information panel of the camera body is turned on. When the information panel is turned off, the value of this capability is set to OFF (False).

### 3.83. FocalLength

This will get the focal length of the lens.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_FocalLength |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Float |
| **ulOperations** | kNkMAIDCapOperation_Get |
| **Data** | lfValue (mm) |

When a CPU lens is not attached, the value of this capability is set to zero.

### 3.84. FocusMode

This will get the focus mode.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_FocusMode |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get |
| **Data** | one of eNkMAIDFocusMode<br>0: MF<br>1: AF-S<br>2: AF-C<br>3: AF-A<br>4: AF-F |

When the lens is not attached, the value of this capability is always MF.

### 3.85. BracketingCount

This will get the number of shots on AE bracketing or ADL bracketing.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_BracketingCount |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get |
| **Data** | [AE Bracketing] 1 – 3<br>[ADL Bracketing] 1 – 2 |

When the Capability_EnableBracketing is true and the value of Capability_BracketingVary is either “AE bracketing” or “ADL bracketing”, this capability is valid. If this capability is invalid, returns zero.

### 3.86. InternalFlashStatus

This will show the status of Built-in flash.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_InternalFlashStatus |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get |
| **Data** | one of eNkMAIDInternalFlashStatus<br>0: Ready 1:Not Ready 2: Close |

### 3.87. InternalFlashComp

This will set the flash compensation of Built-in flash.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_InternalFlashComp |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Range |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set |
| **Data** | -3～+1 (Default:0) |

The module sets the same step value as the value of Capability_EVInterva.

If any of the following, this Capability cannot be set into.

- When Capability_ExposureMode is set to “Auto” , “Auto (flash off)” or Special Effects Modes.
- During movie recording.
- When the value of Capability_HDRMode is except 0 (Off).
- When the Capability_InternalFlashStatus is “Close” and Capability_ExternalFlashStatus is “Not Exist”.
- When Capability_InternalSplMode is “Manual”, Capability_ExternalFlashStatus is “2: Not Exist” and Capability_InternalFlashStatus is expect “2: Close”.
- When Capability_InternalSplMode is “Manual” and Capability_ExternalFlashSort is “4: new communication”.

### 3.88. ExternalFlashStatus

This will shows the status of External flash.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_ExternalFlashStatus |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get |
| **Data** | one of eNkMAIDExternalFlashStatus<br>0: Ready　1: Not Ready　2: Not Exist |

### 3.89. ExternalFlashComp

This will set the flash compensation of the external speedlight.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_ExternalFlashComp |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Range |
| **ulOperations** | kNkMAIDCapOperation_Get |
| **Data** | -3～+3EV (1/6EV step) |

This capability is valid when Capability_ExternalNewTypeFlashMode is either iTTL-BL(1), iTTL(2), AA(3) or GN(5).

### 3.90. ExternalFlashSort

This will get the sort of external speedlight.

**Capability** kNkMAIDCapability_ExternalFlashSort

**Object types** Source

**ulType** kNkMAIDCapType_Unsigned

**ulOperations** kNkMAIDCapOperation_Get

**Data** one of eNkMAIDExternalFlashSort

| |
|---|
| 0: non- communication. |
| 2: new communication (with setting display) |
| 4: new communication (without setting display:SB-400). |
| 3: Not exist. |

The camera cannot detect “1: old communication.”, so this capability returns always “0: non- communication.” when an Old communication speedlight is attached.

The relationship of external speedlight type and the speedlight made by Nikon is shown in the following table.

| New communication (with setting display) | New communication (without setting display) | Old communication | Non- communication | Not detected |
|---|---|---|---|---|
| SB-910<br>SB-900、<br>SB-800、<br>SB-700、<br>SB-600、<br>SU-800 | SB-400<br>SB-300 | SB-80DX、<br>SB-50DX、<br>SB-28DX、<br>SB-28D、<br>SB-28、<br>SB-27、<br>SB-26、<br>SB-25、<br>SB-24、 | SB-30、<br>SB-29、<br>SB-29S、<br>SB-23、<br>SB-22、<br>SB-22S、<br>SB-21A、<br>SB-21B、<br>SB-20、<br>SB-19、<br>SB-18、<br>SB-17、<br>SB-16A、<br>SB-16B、<br>SB-15、<br>SB-14、<br>SB-12、<br>SB-11、<br>SB-10、<br>SB-E | SB-9、<br>SB-8、<br>SB-7、<br>SB-6、<br>SB-5、<br>SB-4、<br>SB-3、<br>SB-2、<br>SB-1 |

### 3.91. ExternalNewTypeFlashMode

This will get flash mode when the Capability_ExternalFlashSort is 2 (new communication (with setting display)) or 4 (new communication (without setting display))

| | |
|---|---|
| **Capability** | kNkMAIDCapability_ExternalNewTypeFlashMode |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Enum<br>kNkMAIDArrayType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, |
| **Data** | one of eNkMAIDExternalNewTypeFlashMode<br>0: OFF<br>1: iTTL-BL<br>2: iTTL<br>3: AA(Auto aperture)<br>4: A(Non-TTL auto)<br>5: GN(Range-priority manual)<br>6: M(manual)<br>7: Repeating flash<br>8: The external speed light, new communication does not exist. |

### 3.92. LensInfo

This will get the focal length and minimum F number.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_LensInfo |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_String |
| **ulOperations** | kNkMAIDCapOperation_Get |
| **Data** | (e.g.)”35-70/F3.3-4.5D” |

In a following lens type case, a corresponding character string is added.

- In the case of VR type lens,”VR”is added to a head.
- In the case of D type lens,”D” is added to an end.
- In the case of G type lens,”G” is added to an end.
- In the case of E type lens,”E” is added to an end.

### 3.93. RetractableLensWarningStatus

This will show the status of RetractableLensWarning.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_RetractableLensWarningStatus |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Boolean |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault |
| **Data** | True: Warning about retraction<br>False: Not warning about retraction |

### 3.94. AFCapture

This will take a picture after auto focus and save an image to specified media.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_AFCapture |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Process |
| **ulOperations** | kNkMAIDCapOperation_Start |

This will take a picture after auto focus. If the Capability_FocusMode is MF (0) or lens is not attached, the camera does shooting immediately without auto focus.

When auto focus failed, whether taking a picture or returning out of focus error, that is depends on the setting of Capability_FocusMode.

When Capability_ShootingMode is CL or CH, Numbers of photo in which a seriography is possible is the smallest number in Capability_ContinuousShootingNum, Capability_RemainContinuousShooting, or the remaining numbers of bracketing shot.

However, when the value of Capability_InternalFlashStatus is “0: Ready” or when the Capability_HDRMode is set up in addition to “0: Off”, if the Capability_ShootingMode is set as continuous shooting mode, the same operation as “SingleFrame Mode” is taken.

When Capability_LiveViewStatus is ON(1), the ulVisibility of this capability is invalid and the ulOperations is set to invalid.

When the module prepared to get a main image, the module issues kNkMAIDEvent_AddChild to source object.

The media saved an image is specified by Capability_SaveMedia. When there is not free space in specified media, this capability returns kMAIDResult_MediaFull. And this capability returns kNkMAIDResult_NoMedia when card is under being formatted or no card is inserted.

If any of the following, the ulOperations of this capability cannot be perfotmed.

- When the value of Capability_LiveViewStatus is 1(ON).
- When the Capability_ShutterSpeed is set as “Time”.
- When the Capability_RetractableLensWarningStatus is “True: Warning about retraction”.

### 3.95. ContrastAF

This will control contrast AF when Live view is executed on Tripod mode.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_ContrastAF |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set |
| | one of eNkMAIDContrastAF |
| | 0x00: start AF (effective only as the Set value) |
| | 0x01: stop AF (effective only as the Set value) |
| | 0x10: AF finishes in focus (effective only as the Get value) |
| | 0x11: AF finishes out of focus (effective only as the Get value) |
| | 0x12: It is operating AF (effective only as the Get value) |

Contrast AF will start when the client set 0x00 (start AF). And the module will return the response without wait for AF finish.

The client can confirm whether contrast AF finish correctly by getting value of this capability, or referring “focus drive state” of “display information” in Live view image. (please refer NkMAIDCapability_GetLiveViewImage)

When the client wants to stop contrast AF, the client will set 0x01 (stop AF). After contrast AF finish, the module returns response.

The following cases, this capability is invalid and this capability is set to read-only.

- When the CPU lens is not attached.
- When the Capability_FocusMode is “MF”.
- When the value of Capability_LiveViewStatus is 0(OFF).
- When the Capability_RetractableLensWarningStatus is true.

### 3.96. PreCapture

This will take a picture for presetting white balance.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_PreCapture |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Process |
| **ulOperations** | kNkMAIDCapOperation_Start |
| **Data** | None |

If any of the following, the ulOperations of this capability cannot be perfotmed.

- When the Capability_LiveViewStatus is 1(ON).
- When the Capability_ShutterSpeed is set as "Time".
- When the Capability_RetractableLensWarningStatus is "True: Warning about retraction".
- When the value of Capability_HDRMode is except 0 (Off).

### 3.97. MFDriveStep

This will set the driving step of lens for adjusting focus position during live view.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_MFDriveStep |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Range |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set |
| **Data** | driving step (Number of pulses) 1 to 32767 |

This capability will save the driving step internally, does not send request for adjusting focus position to camera. Capability_MFDrive will send request for adjusting focus position to camera with this capability value actually.

The following cases, the ulVisibility of this capability is invalid and the ulOperations is set to invalid.

- When the value of Capabilty_FocusMode is "MF" or "AF~F".
- When CPU lens is not attached.
- When the value of Capabilty_LiveViewStatus is "OFF".

### 3.98. MFDrive

This will adjust focus position during live view.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_MFDrive |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Set |
| **Data** | one of eNkMAIDMFDrive |
| | 0: infinity -> close |
| | 1: close -> infinity |

This will send request to adjust focus position with the setting of this capability and the step of Capability_MFDriveStep.The module will return response as soon as the camera starts adjusting manual focus position, the module doesn't wait to finish manual focus driving. If manual focus driving reaches the end of focus area, the module will return kNkMAIDResult_MFDriveEnd.

After this capability is executed correctly, the client can confirm whether manual focus driving finish correctly by getting value of this capability, or referring "focus drive state" of "Display information" in Live view image.

The following cases, the ulVisibility of this capability is invalid and the ulOperations is set to invalid.

- When the value of Capabilty_FocusMode is "MF" or "AF~F".
- When CPUlens is not attached.
- When the value of Capabilty_LiveViewStatus is "OFF".

### 3.99. ContrastAFArea

This will change focus point of contrast AF during live view.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_ContrastAFArea |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Point |
| **ulOperations** | kNkMAIDCapOperation_Set |
| **Data** | struct NkMAIDPoint<br>{<br>SLONG x; -----Coordinates of X axis<br>SLONG y; -----Coordinates of Y axis<br>} |

This capability set the focus point by using x and y of NkMAIDPoint structure.

The value range of x and y is defined by “total size” of “Display information” in Live view image. ( please refer NkMAIDCapability_GetLiveViewImage)

But the range that can be actually set becomes an area where "size of the AF frame" length and breadth size half was subtracted from the length and breadth size of "total size" respectively. When the value that exceeds the range that can be set to x and y is set, the maximum or minimum value will be used as this value.

Even though Capability_FocusMode is 0(MF), or CPU lens is not attached, this capability can be set.

When the Capability_LiveViewStatus is not “1: ON”, this Capability cannot be set into.

### 3.100. CaptureDustImage

This will take a dust off ref photo and saved to specified media.

**Capability** kNkMAIDCapability_CaptureDustImage

**Object types** Source

**ulType** kNkMAIDCapType_Process

**ulOperations** kNkMAIDCapOperation_Start

The format type of dust off ref photo is kNkMAIDFileDataType_NDF.

When the client deletes a dust off ref photo by Capability_DeleteDramImage, the client must use Item ID notified by data parameter of kNkMAIDEvent_AddChild as Capability_CurrentItemID. The media saved an image is specified by Capability_SaveMedia. When there is not free space in specified media, this capability returns kMAIDResult_MediaFull. And this capability returns

The following cases, this Capability cannot be performed.

- When the capability_LiveViewStatus is 1 (ON).
- When CPUlens is not attached.
- When the value of Capabilty_ShutterSpeed is “Time”.
- Capabilty_RetractableLensWarningStatus is “True: Warning about retraction”.

### 3.101. DeleteDramImage

This will delete DRAM image specified by Capability_CurrentItemID.

**Capability** kNkMAIDCapability_DeleteDramImage

**Object types** Source

**ulType** kNkMAIDCapType_Process

**ulOperations** kNkMAIDCapOperation_Start

The DRAM image to be deleted is specified by Capability_CurrentItemID.

This capability execution timing is limited to the following case.

- After issuing kNkMAIDCapability_Acquire for Image Object, and before issuing kNkMAIDCommand_Close.

The client will issue Capability_Acquire for Image object and cancel Capability_Acquire by kNkMAIDCommand_Abort, and set Capability_CurrentItemID and execute this capability, so, the deletion will be completed.

In case of deletion of RAW+JPEG, if the client executes this capability for JPEG, the both of RAW and Jpeg files will be deleted at the same time.

When the client deletes DRAM image after receiving kNkMAIDEvent_AddChild, the

client must close Item objcet. The module does not close Item object.

This capability is not supported when an image is saved on Crad.

To Delete SDRAM image by this capability is prohibited, when it is applied for either of condition below. In this case, this capability returns kNkMAIDResult_NotSupported.

- When the value of Capability_SaveMedia is “2: Card + SDRAM”.
- When during movie file recording.

### 3.102. RawJpegImageStatus

This will get whether the image is taken on RAW+JPEG mode.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_RawJpegImageStatus |
| **Object types** | Image |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get |
| **Data** | one of eNkMAIDRawJpegImageStatus<br>0: Single 1: Raw+JPEG |

### 3.103. CurrentItemID

This will specify the DRAM image operated now.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_CurrentItemID |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set |

Item ID is used as an identifier that specifies the image data in SDRAM.

Item ID is notified by data parameter of kNkMAIDEvent_AddChild.

The value of this capability is referred by Capability_DeleteDramImage.

### 3.104. GetLiveViewImage

This will get Live view image.

**Capability** kNkMAIDCapability_GetLiveViewImage

**Object types** Source

**ulType** kNkMAIDCapType_Array

kNkMAIDArrayType_Unsigned

**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray

The client will get the size of Live view image by kNkMAIDCapOperation_Get, and get a actual Live view data by kNkMAIDCapOperation_GetArray.

Every time, the size of Live view image need not be confirmed with kNkMAIDCapOperation_Get in this capability before execution of kNkMAIDCapOperation_GetArray because the specification of Live view image is always fixation.

When the client want to get Live view image with kNkMAIDCapOperation_GetArray, the client must allocate the buffer for the maximum size, and set buffer to kNkMAIDArray.pData, and set allocate size to kNkMAIDArray.ulElements.

After reading preview image, kNkMAIDArray.ulElements will be updated with the actual size of Live view image I, the Live view image will be set to kNkMAIDArray.pData.

When Capability_LiveViewStatus is OFF(0), the ulOperations of this capability is set to read-only, kNkMAIDCapOperation_GetArray is invalid.

If Live view is stopped by camera automatically (including when the live view time limit passes), the module returns kNkMAIDResult_NotLiveView.

Live view image is consisted of “Display information” and “Live view image(JPEG).”

The pixel size of Live view image is different in each Live view data, each detailed information is set to “Display information” area.

Specification of Live view image

| image quality | maximum file size |
|---|---|
| Jpeg Basic | Header size 8 byte<br>376 byte / Display information<br>49,920Bbyte / Max Live view image |

The format of the Live view image is shown below.

| | | | | |
|---|---|---|---|---|
| | Display information area size | | 4Byte | |
| | Live view image area size | | 4Byte | |
| Display information | Attached JPEG image size | Horizontal size | 2Byte | JPEG image size is the size that has been set in kNkMAIDCapability_LiveViewImageSize. If during movie recording will change the image size by setting kNkMAIDCapability_MovieScreenSize and kNkMAIDCapability_LiveViewImageSize. |
| | | Vertical size | 2Byte | |
| | Whole size | Horizontal size | 2Byte | Standard of the coordinates |
| | | Vertical size | 2Byte | |
| | Display area size | Horizontal size | 2Byte | The whole size is equal to the display area size when the image is not enlarged. |
| | | Vertical size | 2Byte | |
| | Display center coordinates | Horizontal size | 2Byte | |
| | | Vertical size | 2Byte | |
| | AF frame size | Horizontal size | 2Byte | |
| | | Vertical size | 2Byte | |
| | AF frame center coordinates | Horizontal size | 2Byte | |
| | | Vertical size | 2Byte | |
| | Reserve | | 4Byte | |
| | Reserve | | 1Byte | |
| | Rotation direction | | 1Byte | 0: No rotation<br>1: Rotate counterclockwise<br>2: Rotate clockwise |
| | Focus driving status | | 1Byte | 0: Not driving, 1: Driving |
| | Reserve | | 1Byte | |
| | Reserve | | 4Byte | |
| | Reserve | | 2Byte | |
| | Countdown time | | 2Byte | Countdown every one second starting from 3600 (one hour) ;<br>countdown starting from thirty seconds with a rise in temperature |
| | Focusing judgment result | | 1Byte | 0: No information,<br>1: Not focused,<br>2: Focused |
| | AF driving enabled status | | 1Byte | 0: AF driving disabled,<br>1: AF driving enable |
| | Reserve | | 2Byte | |

| | | | | | |
|---|---|---|---|---|---|
| | Reserve | | | 12Byte | Fixed to 0 for D5500 |
| | Remaining time of movie recording | | | 4Byte | From 0 to 1200000 [msec]<br>* It is valid during the movie<br>recording state. |
| | Movie recording information | | | 1Byte | 0: During LV execution<br>1: During movie recording |
| | AF mode status of the face detection system | | | 1Byte | 0: The face detectionAF is OFF<br>1: The face detection AF is ON |
| | The number of persons whose faces are detected by the system | | | 1Byte | From 0 to 35 (Thirty-five is the maximum number of persons for D5500) |
| | AF area index | | | 1Byte | From 0 to 34<br>(fixed to 0 for D5500) |
| | 0<br>~<br>34 | AF frame size | Horizonal size | 2Byte | Area of the AF frame size and the AF<br>frame center coordinates for thirty-five persons<br>(4 Byte + 4 Byte) x 35 persons;<br>280 Byte in total |
| | | | Vertical size | 2Byte | |
| | | AF frame center coordinates | Horizonal position | 2Byte | |
| | | | Vertical position | 2Byte | |
| | Sound level (peak) | | L | 1Byte | 0～14 |
| | | | R | 1Byte | 0～14 |
| | Sound level (current value) | | L | 1Byte | 0～14 |
| | | | R | 1Byte | 0～14 |
| | Reserve | | | 1Byte | |
| | Reserve | | | 1Byte | |
| | Reserve | | | 1Byte | |
| | Reserve | | | 1Byte | |
| | Reserve | | | 24Byte | |
| Live view image | Image data | | | | |

### 3.105. GetVideoImage

This will get Movie image.

**Capability** kNkMAIDCapability_GetVideoImage

**Object types** Video

**ulType** kNkMAIDCapType_Generic

**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray

**Data** pointer to NkMAIDGetVideoImage structure

```
typedef struct tagNkMAIDGetVideoImage
{
ULONG   ulType;------one of eNkMAIDArrayType
ULONG   ulOffset;-----Offset position that begins data acquisition
ULONG   ulReadSize;-----Size of acquired data
ULONG   ulDataSize;-----Size of buffer set to “pData”
LPVOID   pData;---------Pointer to buffer
} NkMAIDGetVideoImage, FAR* LPNkMAIDGetVideoImage;
```

The client will get the size of Movie image by kNkMAIDCapOperation_Get, and get an actual Movie data by kNkMAIDCapOperation_GetArray.

While getting movie data, camera is automatically locked state, and the operation of the camera body is impossible.

"While getting movie data" means, it is the period from the first issued kNkMAIDCapOperation_GetArray, until canceled or until the completion of the all movie data.

[In case of Get]

The data size for the unacquisition is set to kNkMAIDGetVideoImage.ulDataSize.

[In case of GetArray]

When the client want to get Movie image with kNkMAIDCapOperation_GetArray, the client must allocate the buffer for size to be acquired, and set buffer to kNkMAIDGetVideoImage.pData, and set allocate size to kNkMAIDGetVideoImage.ulElements, and set offset position to kNkMAIDGetVideoImage.ulOffset.

After reading, the size of data actually read will be set to kNkMAIDGetVideoImage.ulReadSize and the Movie image will be set to kNkMAIDGetVideoImage.pData.

It is necessary to set “kNkMAIDArrayType_Unsigned” to kNkMAIDGetVideoImage.ulType. When the value that exceeds the size of actual movie data is set, module returns kNkMAIDResult_ValueOutOfBounds.

[Cancel of getting movie data]

To cancel getting movie data, call kNkMAIDCapOperation_GetArray set to 0 to ulDataSize.

In the following cases, getting movie data will be canceled automatically by the camera.

- When the interval is issued kNkMAIDCapOperation_GetArray exceeds about 60 seconds.
- Doing the following operations when "While getting movie data"
    1. Called the Capability other than GetVideoImage.
    2. Inserting or removing the card.

### 3.106. LockCamera

This will lock camera. When the camera is locked, user can't operate it directly.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_LockCamera |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Boolean |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set<br>kNkMAIDCapOperation_GetDefault |
| **Data** | True: Lock False: Unlock |

When Capability_LiveViewStatus is 1(ON), the ulOperations of this capability is set to read-only.

### 3.107. CameraType

This will get the camera type.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_CameraType |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get |
| **Data** | one of eNkMAIDCameraType<br>0x3C: D5500<br>0x44: D5600 |

### 3.108. LensType

This will get the lens type about CPU lens.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_LensType |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get |
| **Data** | one of eNkMAIDLensType |
| | 0x00000001: D type |
| | 0x00000010: G type |
| | 0x00000100: VR |
| | 0x00001000: DX |
| | 0x00010000: AF-S |
| | 0x00100000: Auto distortion control |
| | 0x01000000: Retractable lens |
| | 0x00000020: E type (Electromagnetism diaphragm) |
| | 0x00000040: STM lens |

The value of this capability is expressed by the OR value.

When CPU lens is not attached, the module returns 0.

### 3.109. AFMode

This will set the focus mode for phase detection on still image shooting.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_AFMode |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault |
| | kNkMAIDCapOperation_Set |
| **Data** | one of eNkMAIDAFMode |
| | 0: AF-S |
| | 1: AF-C |
| | 2: AF-A |
| | 3: MF Fixed |
| | 4: MF selected |

This capability is affected by setting of Capability_FocusMode on still image shooting, Capability_LensType, Capability_ExposureMode.

When there is a change in the types of values that can be set in this Capability, the module sends to the client kMAIDEvent_CapChange.

The following cases, the ulOperations of this capability is set to read-only.

- Capability_AFMode is MF Fixed(3).
- AF-S lens is not attached.
- Capability_ExposureMode is Special Effects Modes (because Capability_AFMode is set to MF fixed(3))
- When the ulOperations of Capability_LiveViewStatus is ON(1)
- When the Capability_RetractableLensWarningStatus is "True: Warning about Retraction".

| Conditions | Shooting mode | AFMode |
|---|---|---|
| AF-S lens is not attached. | | MF Fixed |
| AF-S lens is attached＋MF setting (lens settings) | | MF Fixed |
| Retractable lens is attached + RetractableWarning | | MF Fixed |
| AF-S lens is attached＋AF setting (lens settings) | P,S,A,M | AF-S, AF-C, AF-A ,MF selected |
| AF-S lens is attached＋AF setting (lens settings) | Scene modes effects | AF-A ,MF selected |

When the value of Capability_FocusAreaMode is "3D-tracking" or "Dynamic(9/21/39points)", if the value of this capability is set to AF-S(0), the value of Capability_FocusAreaMode is changed into "Single" automatically.

### 3.110. AFModeAtLiveView

This will set the focus mode of the live view.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_AFModeAtLiveView |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Enum |
| | kNkMAIDArrayType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, |
| | kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set |
| **Data** | one of eNkMAIDAFModeAtLiveView |
| | 0: AF-S |
| | 2: AF-F |
| | 3: MF fixed (effective only as the Get value) |
| | 4: MF selected |

When Capability_ExposureMode is “Miniature”, “Photo Illustration”, “Toy Camera Effect” of Special Effects modes, AF-F does not enumerate.

If any of the following, the ulOperations cannot be set into.

- This capability is MF fixed(3).
- During movie recording.
- Capability_RetractableLensWarningStatus is “True: Warning about retraction”. (because Capability_AFMode is set to MF fixed(3))
- When the value of Capability_LensType is not “0x00010000: AF-S”.

### 3.111. LiveViewAF

This will set the focus point in live view mode.

**Capability** kNkMAIDCapability_LiveViewAF

**Object types** Source

**ulType** kNkMAIDCapType_Unsigned

**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetDefault, kNkMAIDCapOperation_Set

**Data** one of eNkMAIDLiveViewAF

0: Face priority

1: Wide area

2: Normal area

3: Subject tracking

In the following table, the default value is changed by Scene modes or Special Effects modes.

When the setting of Capability_ExposureMode is changed to Scene Modes or Special Effects modes, the value of this capability will be changed to each default value.

| Capability_ExposureMode<br>Capability_SceneMode<br>Capability_EffectMode | default value |
|---|---|
| Portrait(SceneModes)<br>Landscape(SceneModes)<br>Party/Indoor(SCENE)<br>Beach/Snow(SCENE)<br>Sunset(SCENE)<br>Dusk/Dawn(SCENE)<br>Candlelight(SCENE)<br>Blossom(SCENE)<br>Autumn Colors(SCENE)<br>Night Portrait(SCENE)<br>Child(SceneModes) | 0 : Face priority |
| Auto(Auto/Portrait/<br>Landscape/Night Portrait*1)<br>Flash Off (Auto/Portrait/<br>Landscape/Night Portrait*1) | 0 : Face priority<br>(unchangeable) |
| Sports(SceneModes)<br>Night Landscape(SCENE)<br>Pet Portrait(SCENE)<br>Silhouette(EFFECTS)<br>High Key(EFFECTS)<br>Low Key(EFFECTS)<br>Color Sketch(EFFECTS)<br>Selective Color(EFFECTS)<br>Night Vision(EFFECTS)<br>Toy Camera(EFFECTS)<br>Super Vivid (EFFECTS) | 1 : Wide area |

| | |
|---|---|
| Pop(EFFECTS)<br>Photo Illustration(EFFECTS) | |
| Miniature Effect(EFFECTS) | 1 : Wide area<br>(unchangeable) |
| Close Up(SceneModes)<br>Food(SCENE) | 2 : Normal area |
| Auto(Close Up*1)<br>Flash Off(Close Up*1) | 2 : Normal area<br>(unchangeable) |

*1: The value in parenthesis after “Auto” and “Flash Off” show the value of Capability_AutoSceneModeStatus.

When [3 : Subject tracking] is set while executing a live view, kNkMAIDResult_ValueOutOfBounds is returned.

And, when a live view is begun when [3 : Subject tracking] is set, the value of this capability is automatically changed to [1 : Wide area].

If any of the following, the ulOperations cannot be set into.

- The Capability_ExposureMode is “Auto”, “Flash Off” or “Miniature” of Special Effects modes.
- During movie file recording.

### 3.112. MovRecInCardStatus

This will start or stop movie recording and show status of movie recording.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_MovRecInCardStatus |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault |
| **Data** | one of eNkMAIDMovRecInCardStatus<br>0: OFF<br>1: ON |

When the client start movie recording in the card, the client must set the value of this capability to ON(1). And when the client stop movie recording in the card, the client must set the value of this capability to OFF(0).

In case of kNkMAIDCapOperation_Get, the value of this capability will show the current status of movie recording.

This capability is accepted only during Live view execution.

It is recommended to check the value of Capability_MovRecInCardProhibit before issuing this capability. If the Capability_MovRecInCardProhibit is a value other than 0, the client cannot start movie recording.

When the Capability_LiveViewStatus is set to OFF(0), movie recording is automatically stopped by the camera.

Taking a picture of the still picture is prohibited while movie file recording.

The following cases, this Capability cannot be performed.

- When the Capability_LiveViewStatus is “OFF(0)”.

### 3.113. MovRecInCardProhibit

This will show the status of movie recording prohibition.

**Capability** kNkMAIDCapability_MovRecInCardProhibit

**Object types** Source

**ulType** kNkMAIDCapType_Unsigned

**ulOperations** kNkMAIDCapOperation_Get

**Data** one of eNkMAIDMovRecInCardProhibit

The movie recording prohibition is shown by the OR value of the following definition value. When 0 returns, the status is not movie recording prohibition.

| Value | Prohibition condition |
|---|---|
| 0x00001000 | During enlarged display of Live view |
| 0x00000800 | Card protected |
| 0x00000400 | During movie file recording |
| 0x00000200 | There is unrecorded image or movie data in the buffer. |
| 0x00000008 | No free area in the card |
| 0x00000004 | Card not formatted |
| 0x00000002 | Card error |
| 0x00000001 | No card inserted |

This Capability becomes effective only while executing a live view.

When the Capability_LiveViewStatus is OFF(0), the value of this capability is not fixed. Even if a value has been entered, it is not guaranteed.

### 3.114. SaveMedia

This will set the recording media by shooting, shutter-release button on camera body or Capability_Capture or Capability_AFCapture, Capability_CaptureDustImage.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_SaveMedia |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault |
| **Data** | one of eNkMAIDSaveMedia<br>0: Card<br>1: SDRAM<br>2: Card + SDRAM |

When during movie file recording, this Capability cannot be set into.

### 3.115. BlinkingStatus

This will show the status of the display a shutter speed and an aperture of the camera.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_BlinkingStatus |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get |
| **Data** | one of eNkMAIDBlinkStatus<br>0: Both a shutter speed and an aperture are displayed with normal status.<br>1: Only a shutter speed is displayed with blinking status.<br>2: Only an aperture is displayed with blinking status.<br>3: Both a shutter speed and an aperture are displayed with blinking status. |

In case of bulb or warning of the time, the value of this Capability is set to 1(flashing only the shutter speed).

### 3.116. AutoSceneModeStatus

This will show the scene mode which the Automatic Scene Selection selects automatically.

**Capability** kNkMAIDCapability_AutoSceneModeStatus

**Object types** Source

**ulType** kNkMAIDCapType_Enum

kNkMAIDArrayType_PackedString

**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray,

**Data**

| | |
|---|---|
| Automatic Scene Selection is invalid. | "Unjudgment" |
| Auto | "Auto" |
| Portrait | "Portrait" |
| Landscape | "Landscape" |
| Close up | "Closeup" |
| Night portrait | "NightPortrait " |

This capability is valid if the Capability_ExposureMode is either "Auto" or "Flash off" when Capability_LiveViewStatus is ON(1). Other than above conditions, Automatic Scene Selection is invalid, so the value of this capability is set to "Unjudgment".

### 3.117. ISOControlSensitivity

This capability will show ISO sensitivity which the camera is controlling.

**Capability** kNkMAIDCapability_ISOControlSensitivity

**Object types** Source

**ulType** kNkMAIDCapType_Unsigned

**ulOperations** kNkMAIDCapOperation_Get

**Data** eNkMAIDISOControlSensitivity2

| value | ISOControlSensitivity | value | ISOControlSensitivity |
|---|---|---|---|
| 100 | ISO100 | 3200 | ISO3200 |
| 110 | ISO110 | 3600 | ISO3600 |
| 125 | ISO125 | 4000 | ISO4000 |
| 140 | ISO140 | 4500 | ISO4500 |
| 160 | ISO160 | 5000 | ISO5000 |
| 180 | ISO180 | 5600 | ISO5600 |
| 200 | ISO200 | 6400 | ISO6400 |
| 220 | ISO220 | 7200 | ISO7200 |
| 250 | ISO250 | 8000 | ISO8000 |
| 280 | ISO280 | 9000 | ISO9000 |
| 320 | ISO320 | 10000 | ISO10000 |
| 360 | ISO360 | 11000 | ISO11000 |
| 400 | ISO400 | 12800 | ISO12800 |
| 450 | ISO450 | 14400 | ISO14400 |
| 500 | ISO500 | 16000 | ISO16000 |
| 560 | ISO560 | 18000 | ISO18000 |
| 640 | ISO640 | 20000 | ISO20000 |
| 720 | ISO720 | 22000 | ISO22000 |
| 800 | ISO800 | 25600 | ISO25600 |
| 900 | ISO900 | 28800 | Hi 0.2 |
| 1000 | ISO1000 | 32000 | Hi 0.3 |
| 1100 | ISO1100 | 36000 | Hi 0.5 |
| 1250 | ISO1250 | 40000 | Hi 0.7 |
| 1400 | ISO1400 | 45600 | Hi 0.8 |
| 1600 | ISO1600 | 51200 | Hi 1.0 |
| 1800 | ISO1800 | 57600 | Hi 1.2 |
| 2000 | ISO2000 | 64000 | Hi 1.3 |
| 2200 | ISO2200 | 72000 | Hi 1.5 |
| 2500 | ISO2500 | 81200 | Hi 1.7 |
| 2800 | ISO2800 | 91200 | Hi 1.8 |
| - | - | 102400 | Hi 2.0 |

### 3.118. TerminateCapture

This capability will terminate bulb exposure shooting and record to SDRAM the image until stopped bulb.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_TerminateCapture |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_Generic |
| **ulOperations** | kNkMAIDCapOperation_Start |
| **Data** | pointer to NkMAIDTerminateCapture structure |
| | typedef struct tagNkMAIDTerminateCapture |
| | { |
| | ULONG ulParameter1; ---------- Set to 0 |
| | ULONG ulParameter2; ---------- Set to 0 |
| | } NkMAIDTerminateCapture, FAR* LPNkMAIDTerminateCapture; |

The following cases, this capability cannot be performed.

- During movie file recording.
- When the Capability_ExposureMode is not “3: Manual”.
- When the Capability_ShutterSpeed is not “Bulb”.

### 3.119. RawJpegTransferStatus

This capability will set the type of SDRAM image which be notified and transmitted to a client, when the value of Capability_SaveMedia is “2: Card + SDRAM” and the value of Capability_CompressionLevel is “RAW + JPEG”.

| | |
|---|---|
| **Capability** | kNkMAIDCapability_RawJpegTransferStatus |
| **Object types** | Source |
| **ulType** | kNkMAIDCapType_ Boolean |
| **ulOperations** | kNkMAIDCapOperation_Get, |
| | kNkMAIDCapOperation_Set, kNkMAIDCapOperation_GetDefault |
| **Data** | True: JPEG only |
| | False: RAW & JPEG |

The following cases, the ulVisibility of this capability is invalid. (The ulOperations can set.)

- When the value of Capability_CompressionLevel is except “RAW + JPEG”.
- When the value of Capability_SaveMedia is except “2: Card + SDRAM”.

This capability is always can be set into but setting change of this capability is taken as prohibition until it receives kNkMAIDEvent_CaptureComplete about SDRAM from the execution start of “Capability_Capture”, “Capability_AFCapture”, and “Capability_CaptureDustImage”

# 4. Standard Capabilities

## 4.1. AsyncRate

| | |
|---|---|
| **Capability** | kNkMAIDCapability_AsyncRate |
| **Object types** | Module |
| **ulType** | kNkMAIDArrayType_Unsigned |
| **ulOperations** | kNkMAIDCapOperation_Get |

## 4.2. ProgressProc

| | |
|---|---|
| **Capability** | kNkMAIDCapability_ProgressProc |
| **Object types** | Source, Image, Thumbnail, Video |
| **ulType** | kNkMAIDCapType_Callback |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set |

## 4.3. EventProc

| | |
|---|---|
| **Capability** | kNkMAIDCapability_EventProc |
| **Object types** | Module, Source, Item, Image, Thumbnail, Video |
| **ulType** | kNkMAIDCapType_Callback |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set |

## 4.4. DataProc

| | |
|---|---|
| **Capability** | kNkMAIDCapability_DataProc |
| **Object types** | Image, Thumbnail |
| **ulType** | kNkMAIDCapType_Callback |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set |

## 4.5. UIRequestProc

| | |
|---|---|
| **Capability** | kNkMAIDCapability_UIRequestProc |
| **Object types** | Module |
| **ulType** | kNkMAIDCapType_Callback |
| **ulOperations** | kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set |

### 4.6. IsAlive

**Capability** kNkMAIDCapability_IsAlive
**Object types** Module, Source, Item, Image, Thumbnail, Video
**ulType** kNkMAIDCapType_Boolean
**ulOperations** kNkMAIDCapOperation_Get

### 4.7. Children

**Capability** kNkMAIDCapability_Children
**Object types** Module, Source
**ulType** kNkMAIDCapType_Enum
kNkMAIDArrayType_Unsigned
**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray

### 4.8. State

**Capability** kNkMAIDCapability_State

Not supported

### 4.9. Name

**Capability** kNkMAIDCapability_Name
**Object types** Module, Source, Item, Image, Thumbnail, Video
**ulType** kNkMAIDCapType_String
**ulOperations** kNkMAIDCapOperation_Get

The image saved on SDRAM is taken a picture on the mode Capability_SaveMedia is "1:SDRAM", the value of this capability about Item、Image、Thumbnail is "DSC_0000.xxx".

The image saved on SDRAM is taken a picture on the mode Capability_SaveMedia is "2:Card + SDRAM", the value of this capability about Item、Image、Thumbnail is "folder name¥filename.xxx". However, when the image doesn't exist the on the card (For the reasons card was not inserted), "DSC_0000.xxx" is used.

### 4.10. Description

**Capability** kNkMAIDCapability_Description

Not supported

### 4.11. Interface

**Capability** kNkMAIDCapability_Interface
**Object types** Source
**ulType** kNkMAIDCapType_String
**ulOperations** kNkMAIDCapOperation_Get

### 4.12. DataTypes

**Capability** kNkMAIDCapability_DataTypes
**Object types** Source, Item
**ulType** kNkMAIDCapType_Unsigned
**ulOperations** kNkMAIDCapOperation_Get

### 4.13. DateTime

**Capability** kNkMAIDCapability_DateTime
**Object types** Item
**ulType** kNkMAIDCapType_DateTime
**ulOperations** kNkMAIDCapOperation_Get

### 4.14. StoredBytes

**Capability** kNkMAIDCapability_StoredBytes
**Object types** Item, Image, Thumbnail, Video
**ulType** kNkMAIDCapType_Unsigned
**ulOperations** kNkMAIDCapOperation_Get

### 4.15. Eject

**Capability** kNkMAIDCapability_Eject
Not supported

### 4.16. Feed

**Capability** kNkMAIDCapability_Feed
Not supported

### 4.17. Capture

This will take a picture and save the image to specified media.

**Capability** kNkMAIDCapability_Capture

**Object types** Source

**ulType** kNkMAIDCapType_Process

**ulOperations** kNkMAIDCapOperation_Start

When the Capability_ShootingMode is 1(C) or 2(CH), the number of shots set by the Capability_ContinuousShootingNum is taken on continuous shooting mode.

Number of shots is the smallest in value of the Capability_ContinuousShootingNum or value of the Capability_RemainContinuousShooting or Bracketing Shot is remaining number of frames.

When the Capability_ShootingMode is 4(Mirror up), or movie file recording and Capability_MovieReleaseButton is not Live frame grab, the ulVisibility and ulOperations of this capability is set to invalid.

When image is prepared to acquire, kNkMAIDEvent_Add is issued to source object.

If the client executes this capability while doing Live view, Live view will be stopped by camera, and the camera take a picture with AF position set on Live view without Auto focus.

The media saved an image is specified by Capability_SaveMedia. When there is not free space in specified media, this capability returns kMAIDResult_MediaFull. And this capability returns kNkMAIDResult_NoMedia when card is under being formatted or no card is inserted. This Capability becomes invalid during movie recording.

Bulb exposure shootng can be executed when Capability_ExposureMode is Manual and Capability_ShutterSpeed is buib. The bulb shooting is started by this capability and terminated by Capability_TerminateCapture.

When live frame grab is active, if the Capability_ShootingMode is 1(C) or 2(CH), shooting is possible only one picture.

The following cases, the ulVisibility of this capability is invalid and the ulOperations is set to invalid.

- When the value of Capabilty_ShutterSpeed is “Time”.
- When the value of Capabilty_ RetractableLensWarningStatus is “True”.
- When during movie recording.
- Capability_ShootingMode is Mirror up.

### 4.18. Mode

**Capability** kNkMAIDCapability_Mode

Not supported

### 4.19. Acquire

**Capability** kNkMAIDCapability_Acquire

**Object types** Image, Thumbnail

**ulType** kNkMAIDCapType_Process

**ulOperations** kNkMAIDCapOperation_Start

When Object types are set to Thumbnail, this Capability may return kNkMAIDResult_NotSupported error.

When it detects the camera's internal image generation, the module starts reading the Image. Then, the module cache inside it.(Look-ahead processing)

Thumbnail obtaining and caching is not performed in the look-ahead processing.

When loading is complete Image, Inside the camera, the entire image data including a Thumbnail will be deleted.

Therefore, if after the completion of the process ahead, was executed this Capability for Thumbnail, this Capability is kNkMAIDResult_NotSupported error.

For reacquisition when the shooting took place at set "RAW + JPEG". If one image data can be transmitted is successfully carried out only image data of the other re-transmission.

### 4.20. Start

**Capability** kNkMAIDCapability_Start

Not supported

### 4.21. Length

**Capability** kNkMAIDCapability_Length

Not supported

### 4.22. SampleRate

**Capability** kNkMAIDCapability_SampleRate

Not supported

### 4.23. Stereo

**Capability** kNkMAIDCapability_Stereo

Not supported

### 4.24. Samples

**Capability** kNkMAIDCapability_Samples

Not supported

### 4.25. Filter

**Capability** kNkMAIDCapability_Filter

Not supported

### 4.26. Prescan

**Capability** kNkMAIDCapability_Prescan

Not supported

### 4.27. AutoFocus

This will execute phase detection AF.

**Capability** kNkMAIDCapability_AutoFocus

**Object types** Source

**ulType** kNkMAIDCapType_Process

**ulOperations** kNkMAIDCapOperation_Start

The following cases, the ulVisibility of this capability is invalid and the ulOperations is set to invalid.

- When the value of Capability_SpotWBMode is “0(OFF)”.
- When the value of Capabilty_FocusMode is “MF”.
- When CPUlens is not attached.
- When the value of Capability_LiveViewStatus is 1(ON).

### 4.28. AutoFocusPt

**Capability** kNkMAIDCapability_AutoFocusPt

Not supported

### 4.29. Focus

**Capability** kNkMAIDCapability_Focus

Not supported

### 4.30. Coords

**Capability** kNkMAIDCapability_Coords

Not supported

### 4.31. Resolution

**Capability** kNkMAIDCapability_Resolution

Not supported

### 4.32. Preview

**Capability** kNkMAIDCapability_Preview

Not supported

### 4.33. Negative

**Capability** kNkMAIDCapability_Negative

Not supported

### 4.34. Bits

**Capability** kNkMAIDCapability_Bits

Not supported

### 4.35. Planar

**Capability** kNkMAIDCapability_Planar

Not supported

### 4.36. Lut

**Capability** kNkMAIDCapability_Lut

Not supported

### 4.37. Transparency

**Capability** kNkMAIDCapability_Transparency

Not supported

### 4.38. Threshold

**Capability** kNkMAIDCapability_Threshold

Not supported

### 4.39. Pixels

**Capability** kNkMAIDCapability_Pixels

**Object types** Image, Thumbnail, Video

**ulType** kNkMAIDCapType_Size

**ulOperations** kNkMAIDCapOperation_Get

### 4.40. ForceScan

**Capability** kNkMAIDCapability_ForceScan

Not supported

### 4.41. ForcePrescan

**Capability** kNkMAIDCapability_ForcePrescan

Not supported

### 4.42. ForceAutoFocus

**Capability** kNkMAIDCapability_ForceAutoFocus

Not supported

### 4.43. NegativeDefault

**Capability** kNkMAIDCapability_NegativeDefault

Not supported

### 4.44. Firmware

**Capability** kNkMAIDCapability_Firmware

Not supported

### 4.45. CommunicationLevel1

**Capability** kNkMAIDCapability_CommunicationLevel1

Not supported

### 4.46. CommunicationLevel2

**Capability** kNkMAIDCapability_CommunicationLevel2

Not supported

### 4.47. BatteryLevel

**Capability** kNkMAIDCapability_BatteryLevel

**Object types** Source

**ulType** kNkMAIDCapType_Integer

**ulOperations** kNkMAIDCapOperation_Get

**Data** 1, 5, 20, 35, 100

| Data | Battery Indicator |
|---|---|
| 100 | Battery fully charged. |
| 35 | Battery partially discharged. |
| 20 | Low battery.<br>Ready fully-charged spare battery or prepare to charge battery |
| 5 | Battery exhausted.<br>Shutter release disabled.<br>Charge or exchange battery. |
| 1 | The monitor does not turn on. |

This will show the remains of battery by percent.

When the external power supply is used, this capability returns -1.

### 4.48. FreeBytes

Not supported

### 4.49. FreeItems

Not supported

### 4.50. Remove

Not supported

### 4.51. FlashMode

**Capability** kNkMAIDCapability_FlashMode

**Object types** Source

**ulType** kNkMAIDCapType_Enum

kNkMAIDArrayType_Unsigned

**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_GetArray, kNkMAIDCapOperation_Set,

**Data** one of eNkMAIDFlashMode, eNkMAIDFlashModeDX2

0: FrontCurtain

1: Rear-curtain sync

2: Slow

3: Red-eye reduction

| Flash mode / ExposureMode / SceneMode | Front Curtain | Slow | Rear-curtain sync (Slow rear-curtain sync on P,A) | Red-eye reduction | Slow sync with red-eye reduction | flash off |
|---|---|---|---|---|---|---|
| P, A | ●, △ | ●, △ | ●, △ | ●, △ | ●, △ | — |
| S, M | ●, △ | — | ●, △ | ●, △ | — | — |
| Auto(Auto/Portrait/Close up)<br>Portrait (Scene modes)<br>Close up (Scene modes)<br>Child (Scene modes)<br>Party / Indoor (SCENE)<br>Pet Portrait (SCENE)<br>Toy camera(EFFECTS)<br>Super Vivid(EFFECTS)<br>Pop(EFFECTS)<br>Photo Illustration(EFFECTS) | ●, △ | — | — | ●, △ | — | ● |
| Landscape (Scene modes)<br>Sports (Scene modes)<br>Night Landscape (SCENE)<br>Beach / Snow (SCENE)<br>Sunset (SCENE)<br>Dusk/Dawn (SCENE)<br>Candlelight (SCENE)<br>Blossom (SCENE)<br>Autumn Colors (SCENE)<br>Silhouette (EFFECTS)<br>High Key (EFFECTS)<br>Low Key (EFFECTS)<br>Miniature Effect (EFFECTS)<br>Auto(Landscape) | △ | — | — | △ | — | ● |
| Night Portrait (SCENE)<br>Auto(Night Portrait) | — | ●, △ | — | — | ●, △ | ● |
| Flash Off<br>Selective Color (EFFECTS)<br>Night Vision (EFFECTS) | — | — | — | — | — | ●, △ |

| | | | | | | |
|---|---|---|---|---|---|---|
| Food（SCENE） | ●,△ | — | — | — | — | — |

●: When Internal speed light is active(=external speed light is not active), it is available.

△: When external speed light is active (=external speed light is attached and power on), it is available.

—: Not supported

Words in parenthesis after "Auto" show the value of Capability_AutoSceneModeStatus.

Auto(Auto) is selected when Live view (still picture) is inactive.

When Capability_ExternalNewTypeFlashMode is (7) "Repeating flash" and Capability_ExposureMode is P, S, A, M, 1 "[Rear-curtain sync" can not be set and 0 "FrontCurtain" will be set.

The following cases, this Capability cannot be set into.

- During movie file recording.
- When the value of Capabilty_HDRMode is except "0: Off".
- Choice if only one.
- When the Capability_ExposureMode is set into "Auto (flash off)", "SCENE(Food)" or Special Effects Modes(Night Vision, Selective Color).

### 4.52. ModuleType

**Capability** kNkMAIDCapability_ModuleType
**Object types** Module
**ulType** kNkMAIDCapType_Unsigned
**ulOperations** kNkMAIDCapOperation_Get

### 4.53. AcquireStreamStart

**Capability** kNkMAIDCapability_AcquireStreamStart
Not supported

### 4.54. AcquireStreamStop

**Capability** kNkMAIDCapability_AcquireStreamStop
Not supported

### 4.55. AcceptDiskAcquisition

**Capability** kNkMAIDCapability_AcceptDiskAcquisition
**Object types** Source
**ulType** kNkMAIDCapType_Generic
**ulOperations** kNkMAIDCapOperation_Get, kNkMAIDCapOperation_Set

### 4.56. Version

**Capability** kNkMAIDCapability_Version
**Object types** Module
**ulType** kNkMAIDCapType_Unsigned
**ulOperations** kNkMAIDCapOperation_Get

### 4.57. FilmFormat

**Capability** kNkMAIDCapability_FilmFormat
Not supported

### 4.58. TotalBytes

**Capability** kNkMAIDCapability_TotalBytes
Not supported

## 5. Event

The client can't receive the event as follows while opening item object.

### 5.1. AddChild

This event will be issued when the child is added under an object.

**Event** kNkMAIDEvent_AddChild

**Object types** Module, Source, Item

**data parameter** Added child ID

When the added child is Item Object, Item ID will be set to the data parameter of call back function.

This event is issued only about the image preserved in SDRAM. The image preserved on the card is not issued.

### 5.2. RemoveChild

This event will be issued when the child is removed under an object.

**Event** kNkMAIDEvent_RemoveChild

**Object types** Module, Source, Item

**data parameter** Removed child ID

### 5.3. WarmingUp

**Event** kNkMAIDEvent_WarmingUp

Not supported

### 5.4. WarmedUp

**Event** kNkMAIDEvent_WarmedUp

Not supported

### 5.5. CapChange

This event will be issued when the information of Capability is changed.

**Event** kNkMAIDEvent_CapChange

**Object types** Module, Source, Item

**data parameter** Capability ID

In the following cases, this event will be issued.

- When the content of structure "NkMAIDCapInfo" of capability was changed.

- When the array data of capability with the type of kNkMAIDCapType_Array was changed.

### 5.6. OrphanedChildren

**Event** kNkMAIDEvent_OrphanedChildren

Not supported

### 5.7. CapChangeValueOnly

This event will be issued when the current value of capability is changed.

**Event** kNkMAIDEvent_CapChangeValueOnly

**Object types** Module, Source, Item, Data

**data parameter** Capability ID

This event will be issued when only the current value of capability is changed (the array data, ulVisibility, ulOperations is not changed).

### 5.8. CaptureComplete

This will be issued when the acquisition or deletion of all images of which it takes a picture is completed.

**Event** kNkMAIDEvent_CaptureComplete

**Object types** Source

**data parameter** 0 : The all SDRAM images by Capture, AFCapture, CaptureDustImage are finished to read or deleted.
1: The all images by Capture, AFCapture, CaptureDustImage are finished to record in card.

It shows that the all images are finished to record in card or the all SDRAM images were finished to read or deleted.

This event is not issued in case of shooting by shutter-release button.

### 5.9. AddChildInCard

This event will be issued when the child is added in card.

**Event** kNkMAIDEvent_AddChildInCard

**Object types** Item

**data parameter** Added child ID

When the added child is Item Object, Item ID will be set to the data parameter of call back function.

This event is issued only about the movie data preserved on the card. The still image data preserved on the card is not issued.

### 5.10. RecordingInterrupted

This will be issued the factor of discontinue when the recording of movie was stopped with error occurring.

**Event** kNkMAIDEvent_RecordingInterrupted

**Object types** Source

**data parameter** 1: Some error.

0: Low access speed card error.

### 5.11. CapChangeOperationOnly

Operation value or visibility of Capability has changed.

**Event** kNkMAIDEvent_CapChangeOperationOnly

**Object types** Module, Source, Item, Data

**dataparamater** Capability ID

Events to notify (the number of elements in the array, enumeration value configuration, value, etc. has not changed) if only Operation or visibility · Invalid attribute of Capability has changed.

# 6. Vendor Unique Results

## 6.1. ApertureFEE

The aperture is not set maximum F number.

**Result** kNkMAIDResult_ApertureFEE

**Command** Start

**Capability** Capture, AFCapture, PreCapture, CaptureDustImage

**Explanation** If the ExposureMode is set to Program or SpeedPriority and the aperture ring of the lens is not set to maximum F number, the camera cannot execute capture command.

**Expected Action** The client displays the message to set the aperture to maximum F number and is waiting for next command.

## 6.2. BufferNotReady

This is not used in the current module.

## 6.3. NormalTTL

The speedlight is set TTL mode.

**Result** kNkMAIDResult_NormalTTL

**Command** Start

**Capability** Capture

**Explanation** The camera cannot take a picture when an external speedlight is attached and it is set TTL(measuring through the lens) mode.

**Expected Action** The client displays the message that the camera cannot take a picture and is waiting for next command.

## 6.4. MediaFull

There are neither free spaces that can be recorded on the card nor a free space that can be recorded with built-in the camera SDRAM.

**Result** kNkMAIDResult_MediaFull

**Command** Start

**Capability** Capture, AFCapture, CaptureDustImage

**Explanation** There is no free space at the specified media, so the client can not take a picture.

**Expected Action** The client displays the message that the camera cannot take a picture and is waiting for next command.

### 6.5. InvalidMedia

It shows that the client can not take a picture because recording media is broken.

**Result** kNkMAIDResult_InvalidMedia

**Command** Start

**Capability** Capture, AFCapture, CaptureDustImage

**Explanation** When Capability_SaveMedia is "Card" or "Card + SDRAM" it shows the client can not take a picture because the card is broken.

**Expected Action** The client displays the message that the camera cannot take a picture and is waiting for next command.

### 6.6. EraseFailure

This is not used in the current module.

### 6.7. CameraNotFound

The module did not find a camera on the bus.

**Result** kNkMAIDResult_CameraNotFound

**Command** The commands need access to the camera. (most of the commands for the Source, the Item or the Data object.)

**Explanation** The camera was disconnected. If the client sends Async command to the Module object at intervals, it can tell that the camera is reconnected by AddChild event.

**Expected Action** The client displays the message that the camera was disconnected and is waiting for next command.

### 6.8. BatteryDontWork

The main battery in the camera is used up.

**Result** kNkMAIDResult_BatteryDontWork

**Command** Start

**Capability** Capture, AFCapture, CaptureDustImage, PreCapture

**Explanation** The camera cannot take a picture because of the battery.

**Expected Action** The client displays the message that the camera cannot take a picture and suggests changing battery.

### 6.9. ShutterBulb

The exposure time is set to Bulb.

**Result** kNkMAIDResult_ShutterBulb

**Command** Start

**Capability** Capture, AFCapture, CaptureDustImage

**Explanation** The camera cannot execute capture command if the Capability_ShutterSpeed is set to bulb.

**Expected Action** The client displays the message that the camera cannot take a picture and is waiting for next command.

### 6.10. OutOfFocus

Auto focus operation is failed.

**Result** kNkMAIDResult_OutOfFocus

**Command** Start

**Capability** Capture, AutoFocus, AFCapture, CheckContrastAF

**Explanation** When the Capability_FocusMode is AF-S(1) and auto focus operation is failed, the camera cannot take a picture. Then this error is returned for the start of Capability_Capture or Capability_AFCapture.
In case of the Capability_AutoFocus and Capability_CheckContrastAF, this error will be returned when auto focus is failed.

**Expected Action** The client displays the message that the camera is out of focus and is waiting for next command.

### 6.11. Protected

This is not used in the current module.

### 6.12. FileExists

This is not used in the current module.

### 6.13. SharingViolation

This is not used in the current module.

### 6.14. DataTransFailure

An error occurred while data transference.

**Result** kNkMAIDResult_DataTransFailure

**Command** Start, Async

**Capability** Acquire

**Explanation** If this error occurs while the client read an image from DRAM, it will lose the image.

**Expected Action** The client aborts the data transference.

### 6.15. SessionFailure

The module cannot open source object because the camera cannot open more session.

**Result** kNkMAIDResult_SessionFailure

**Command** Open

**Capability** -

**Explanation** The camera can open 1 session. If the client tries to open more source object, the module returns this error.

**Expected Action** The client displays an error message and is waiting for next command.

### 6.16. FileRemoved

This is not used in the current module.

### 6.17. BusReset

This command was aborted because bus-reset occurred.

**Result** kNkMAIDResult_BusReset

**Command** any command

**Capability** any capability

**Explanation** If bus-reset occurred, the command, which the module is executing, is aborted. Then the module returns this result for the command.

**Expected Action** The client sends the command again.

### 6.18. NonCPULens

This is not used in the current module.

### 6.19. ReleaseButtonPressed

This is not used in the current module.

### 6.20. BatteryExhausted

This is not used in the current module.

### 6.21. CaptureFailure

The camera failed in measuring value for white balance preset data.

**Result** kNkMAIDResult_CaptureFailure

**Command** Start

**Capability** PreCapture

**Explanation** When it fails in white balance measurement(Capability_PreCapture), this error is returned.

**Expected Action** The client displays the message to take a picture again and is waiting for next command.

### 6.22. InvalidString

This is not used in the current module.

### 6.23. NotInitialized

This is not used in the current module.

### 6.24. CaptureDisable

This is not used in the current module.

### 6.25. DeviceBusy

A camera did not receive a command.

**Result** kNkMAIDResult_DeviceBusy

**Command** any command

**Capability** any capability

**Explanation** Since a camera is in the state where the command is not receivable, when it is not able to perform, this error returns.

**Expected Action** This command is sent again or a display of a user interface is returned to the state before command execution.

### 6.26. CaptureDustFailure

The camera failed in taking a dust off ref photo.

**Result** kNkMAIDResult_CaptureDustFailure

**Command** Start

**Capability** CaptureDustImage

**Explanation** When it fails in taking a dust off ref photo(Capability_CaptureDustImage), this error is returned.

**Expected Action** Do nothing.

### 6.27. ICADown

Enumeration of device can not be done correctly because ICA does not work on Mac OS X.

**Result** kNkMAIDResult_ICADown

**Command** EnumChildren

**Capability** Children

**Explanation** This error is returned when enumeration of device can not be done correctly because ICA does not work. This error code is used only on Mac OS X.

**Expected Action** The client aborts the command and capability of device search. The client displays the message that the camera must be powered off and client application needs to restart.

### 6.28. NotLiveView

Live view was automatically stopped by the factor of the camera. (Included the case of that the live view time limit passed.)

**Result** kNkMAIDResult_NotLiveView

**Command** Start, Set

**Capability** GetLiveViewImage

**Explanation** Wen live view was automatically stopped by the factor of the camera (Include the case of that the live view time limit passed.), this error is returned.

**Expected Action** The client displays an error message and is waiting for next command.

### 6.29. MFDriveEnd

The focus position reached the end of focus area in manual focus.

**Result** kNkMAIDResult_MFDriveEnd

**Command** Set

**Capability** MFDrive

**Explanation** When the focus position reached the end of focus area by Capability_MFDrive, this error is returned.

**Expected Action** The client displays an error message and is waiting for next command.

### 6.30. UnformattedMedia

It shows that the client can not take a picture because the card is unformatted.

**Result** kNkMAIDResult_UnformattedMedia

**Command** Start

**Capability** Capture, AFCapture, CaptureDustImage

**Explanation** When Capability_SaveMedia is "Card" or "Card + SDRAM" it shows the client can not take a picture because the card is unformatted.

**Expected Action** The client displays the message that the camera cannot take a picture and is waiting for next command.

### 6.31. MediaReadOnly

It shows that the client can not take a picture because the card is protected.

**Result** kNkMAIDResult_MediaReadOnly

**Command** Start

**Capability** Capture, AFCapture, CaptureDustImage

**Explanation** When Capability_SaveMedia is "Card" or "Card + SDRAM" it shows the client can not take a picture because the card is protected.

**Expected Action** The client displays the message that the camera cannot take a picture and is waiting for next command.

### 6.32. BulbReleaseBusy

It shows during bulb exposure shooting.

**Result** kNkMAIDResult_BulbReleaseBusy

**Command** Start, Set

**Capability** Capture

**Explanation** It shows during bulb exposure shooting

**Expected Action** If bulb exposure shooting started by Capability_Capture,

kNkMAIDResult_BulbReleaseBusy will be returned until bulb exposure shooting is terminated.

### 6.33. DuringUpdate

Is not used in the current module.

## 7. kNkMAIDDataObjType_Video

Capability for which data object type kNkMAIDDataObjType_Video can be used by this module applies to the content described in not the MAID3.1 rule but this document.

## 8. Capability table that can be set during live view photography, movie live view, movie recording.

The following table shows the capabilities that can be set during live view.

The capabilities not shown in the table can not be set during live view and Operation is set to read only.

The fields marked with "*" represent that this capability is read only under certain conditions. (For details, please refer each capability fields.)

| Capability | Live View | Movie file Recording |
|---|---|---|
| ImageSize | ○* | × |
| CompressionLevel | ○* | × |
| WBMode | ○* | × |
| Sensitivity | ○* | × |
| ResetMenuBank | ○ | × |
| WBTuneAuto | ○* | × |
| WBTuneIncandescent | ○* | × |
| WBFluorescentType | ○* | × |
| WBTuneFluorescent | ○* | × |
| WBTuneSunny | ○* | × |
| WBTuneFlash | ○* | × |
| WBTuneShade | ○* | × |
| WBTuneCloudy | ○* | × |
| WBPresetNumber | ○* | × |
| WBPresetData | ○ | × |
| WBGainRed | -- | -- |
| WBGainBlue | -- | -- |
| ImageColorSpace | ○ | × |
| IsoControl | ○* | × |
| NoiseReduction | ○* | × |
| NoiseReductionHighISO | ○* | × |
| CompressRAWBitMode | ○ | × |
| PictureControl | ○* | × |
| PictureControlData | ○ | × |
| PictureControlDataEx | ○ | × |

| | | |
|---|---|---|
| GetPicCtrlInfo | -- | -- |
| DeleteCustomPictureControl | ○ | × |
| Active_D_Lighting | ○* | × |
| ISOAutoShutterTime | ○* | × |
| ISOAutoShutterTimeAutoValue | ○* | × |
| ISOAutoHiLimit | ○* | × |
| MovieScreenSize | ○ | × |
| MovieRecMicrophone | ○ | × |
| MovieRecMicrophoneValue | ○* | × |
| MovieWindNoiseReduction | ○* | × |
| MovieManualSetting | ○ | × |
| MovieImageQuality | ○ | × |
| AutoDistortion | ○* | × |
| HDRMode | ○* | × |
| SceneMode | ○* | × |
| EffectMode | ○* | × |
| VignetteControl | ○ | × |
| AFcPriority | × | × |
| AFAreaPoint | × | × |
| EVInterval | ○ | × |
| BracketingVary | ○* | × |
| InternalSplMode | ○* | × |
| VideoMode | ○ | × |
| UserComment | ○ | × |
| EnableComment | ○ | × |
| EnableCopyright | ○ | × |
| ArtistName | ○ | × |
| CopyrightInfo | ○ | × |
| CameraInclinationMode | ○ | × |
| ClockDateTime | ○ | × |
| ShutterSpeed | ○* | ○* |
| FlexibleProgram | ○* | ○* |
| FocusPreferredArea | × | × |
| Aperture | ○* | ○* |
| MeteringMode | ○* | × |

| | | |
|---|---|---|
| ExposureMode | ○* | × |
| ExposureComp | ○* | ○* |
| ShootingMode | ○* | × |
| ContinuousShootingNum | ○ | × |
| FocusAreaMode | × | × |
| EnableBracketing | ○* | ×* |
| AEBracketingStep | ○* | × |
| WBBracketingStep | ○* | × |
| BracketingType | -- | -- |
| ADLBracketingType | -- | -- |
| LiveViewStatus | ○* | ○ |
| LiveViewProhibit | -- | -- |
| LiveViewImageZoomRate | ○ | × |
| LiveViewImageSize | ○ | × |
| CameraInclination | -- | -- |
| RemainContinuousShooting | -- | -- |
| RemainCountInMedia | -- | -- |
| LockExposure | -- | -- |
| LockFocus | -- | -- |
| ExposureStatus | -- | -- |
| InfoDisplayErrStatus | -- | -- |
| FocalLength | -- | -- |
| FocusMode | -- | -- |
| BracketingCount | -- | -- |
| InternalFlashStatus | -- | -- |
| InternalFlashComp | ○* | × |
| ExternalFlashStatus | -- | -- |
| ExternalFlashComp | -- | -- |
| ExternalFlashSort | -- | -- |
| ExternalNewTypeFlashMode | -- | -- |
| LensInfo | -- | -- |
| RetractableLensWarningStatus | -- | -- |
| AFCapture | × | × |
| ContrastAF | ○* | ○* |
| PreCapture | × | × |

| | | |
|---|---|---|
| MFDriveStep | ○* | ○* |
| MFDrive | ○* | ○* |
| ContrastAFArea | ○ | ○ |
| CaptureDustImage | × | × |
| DeleteDramImage | ○* | × |
| RawJpegImageStatus | -- | -- |
| CurrentItemID | ○ | ○ |
| GetLiveViewImage | -- | -- |
| GetVideoImage | -- | -- |
| LockCamera | × | × |
| CameraType | -- | -- |
| LensType | -- | -- |
| AFMode | × | × |
| AFModeAtLiveView | ○* | × |
| LiveViewAF | ○* | × |
| MovRecInCardStatus | ○ | ○ |
| MovRecInCardProhibit | -- | -- |
| SaveMedia | ○ | × |
| BlinkingStatus | -- | -- |
| AutoSceneModeStatus | -- | -- |
| ISOControlSensitivity | -- | -- |
| TerminateCapture | ○* | × |
| RawJpegTransferStatus | ○ | ○ |
| AsyncRate | -- | -- |
| ProgressProc | ○ | ○ |
| EventProc | ○ | ○ |
| DataProc | ○ | ○ |
| UIRequestProc | ○ | ○ |
| IsAlive | -- | -- |
| Children | -- | -- |
| State | -- | -- |
| Name | -- | -- |
| Description | -- | -- |
| Interface | -- | -- |
| DataTypes | -- | -- |

| | | |
|---|---|---|
| DateTime | -- | -- |
| StoredBytes | -- | -- |
| Eject | -- | -- |
| Feed | -- | -- |
| Capture | ○* | × |
| Mode | -- | -- |
| Acquire | ○ | ○ |
| Start | -- | -- |
| Length | -- | -- |
| SampleRate | -- | -- |
| Stereo | -- | -- |
| Samples | -- | -- |
| Filter | -- | -- |
| Prescan | -- | -- |
| AutoFocus | × | × |
| AutoFocusPt | -- | -- |
| Focus | -- | -- |
| Coords | -- | -- |
| Resolution | -- | -- |
| Preview | -- | -- |
| Capability | -- | -- |
| Negative | -- | -- |
| Bits | -- | -- |
| Planar | -- | -- |
| Lut | -- | -- |
| Transparency | -- | -- |
| Threshold | -- | -- |
| Pixels | -- | -- |
| ForceScan | -- | -- |
| ForcePrescan | -- | -- |
| ForceAutoFocus | -- | -- |
| NegativeDefault | -- | -- |
| Firmware | -- | -- |
| CommunicationLevel1 | -- | -- |
| CommunicationLevel2 | -- | -- |

| | | |
|---|---|---|
| BatteryLevel | -- | -- |
| FreeBytes | -- | -- |
| FreeItems | -- | -- |
| Remove | -- | -- |
| FlashMode | ○* | × |
| ModuleType | -- | -- |
| AcquireStreamStart | -- | -- |
| AcquireStreamStop | -- | -- |
| AcceptDiskAcquisition | -- | -- |
| Version | -- | -- |
| FilmFormat | -- | -- |
| TotalBytes | -- | -- |

## 9. History

- Rev.2.3 October 29, 2021
  - Changed the contents of the ASCII characters.
    - 3.50. UserComment

- Rev.2.2 November 15, 2019
  - Changed the description of the following event.
    - 5.8. CaptureComplete

- Rev.2.1 June 14, 2017
  - Added the description of data setting.
    - 3.118. TerminateCapture
  - Changed the execution condition of bulb shooting.
    - 4.17. Capture

- Rev.2.0 December 9, 2016
  - Added informations about D5600.
    - 2. Supported camera
    - 3.33. MovieScreenSize
    - 3.49. VideoMode
    - 3.55. CameraInclinationMode
    - 3.56. ClockDateTime
    - 3.64. ShootingMode
    - 3.107. CameraType
    - 3.108. LensType
    - 4.51. FlashMode

- Rev.1.1 November 13, 2015
  - Changed the description of the following capability.
    - 3.99. ContrastAFArea

- Rev.1.0 October 31. 2014 First version